

USER MANUAL | V0.1 | 2026-05

forenix-oss

OSINT x Forensics - a step-by-step guide for analysts, examiners, and team leads.

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forenix-oss - User Manual

Version 0.1 | 2026-05

Welcome to forenix-oss. This manual is for the analyst, the forensic examiner, and the team lead who'll be using the platform day to day. We assume zero prior knowledge - by the end of this document you'll understand every feature, the workflow it serves, and the keystrokes that make it fast.

The platform is screenshot-driven. Every figure in this manual is a **real** render against a live demo run by hosted LLMs. Nothing is mocked-up in Figma.

Table of contents

1. Introduction
 2. Before you start
 3. Your first launch
 4. The workspace, in 60 seconds
 5. Workflow 1 - Your first OSINT investigation
 6. Workflow 2 - Bridge findings to a forensic case
 7. Workflow 3 - Evidence + chain of custody
 8. Workflow 4 - Verify the audit chain
 9. Workflow 5 - Generate and read reports
 10. Feature reference - every view
 11. Working with the AI adapters
 12. Keyboard shortcuts + command palette
 13. Tips for new analysts
 14. Troubleshooting
 15. Frequently asked questions
 16. Glossary
-

1. Introduction

1.1 What is forenix-oss

forenix-oss is an open-source platform that fuses two workflows that have always lived in separate tools:

- **OSINT (Open-Source Intelligence)** - discovering, organising and reasoning about a target using publicly-available information.
- **Digital forensics** - taking material that may be entered into evidence, hashing it, tracking every change, and proving the custody chain hasn't been tampered with.

Today, an investigator uses one set of tools to do the first half and a completely different set to do the second half. The handoff is manual. The audit trail is artisanal. forenix-oss makes both halves a single, auditable workflow.

1.2 Who this manual is for

- The **analyst** running OSINT pipelines and capturing findings.
- The **forensic examiner** managing evidence and merge-request reviews.
- The **team lead** auditing the chain of custody before hand-off.
- The **operator** deploying the platform and watching its health.

If you're an engineer extending the platform, see `docs/03-SDS.md` , `docs/06-ARCHITECTURE.md` and `docs/08-API.md` instead.

1.3 What you'll learn

By the end of this manual you'll be able to:

1. Run an end-to-end OSINT pipeline against a target.
2. Open a forensic case from any finding and bring its evidence under chain of custody.
3. Verify the integrity of the audit chain yourself.
4. Use every secondary view (network graph, monitors, AI lab, reports, ...) and know what problem each one solves.
5. Switch between AI adapters (mock, NVIDIA, OpenRouter, ...) and understand the tradeoffs.

2. Before you start

2.1 System requirements

- **OS:** Linux, macOS, or Windows-WSL2.
- **Runtime:** Bun 1.3+ or Node.js 20+.
- **Browser:** any modern Chromium-based browser (Chrome, Edge, Brave, Arc) or recent Firefox / Safari.
- **Optional, for hosted LLMs:** an OpenRouter API key (<https://openrouter.ai>), a NVIDIA NIM API key (<https://build.nvidia.com>), or a local Ollama install.

2.2 Where to get help

- This manual.
- The feature catalogue: `docs/FEATURES.md` .
- The operational runbook: `docs/09-RUNBOOK.md` .
- The GitHub issues tracker on <https://github.com/thunderstornX/forenix-oss>.

2.3 The demo dataset

forenix-oss ships with a seed script that creates a complete demo workspace in one command:

- **3 users** - `Admin` , `Jay Investigator` , `Sam Analyst` .

- **2 investigations:**

- INV-2025-019 - Northwind Holdings (status: complete , bridged to CASE-2025-007 - Operation Sandstone).
- INV-2025-020 - Mira Volkov (status: running / complete after the live demo run).
- **1 seeded case** with 3 evidence items, 2 branches, and 6 commits - plus a second case auto-created when the live demo bridges the second investigation.
- A valid 9-row hash-chained audit log to start with; the chain grows naturally as you operate.

Every screenshot in this manual was taken **after** running the live demo, so the numbers and findings you see are the real output of hosted Large Language Models. Your numbers will look slightly different - the model returns slightly different findings on every run, by design.

3. Your first launch

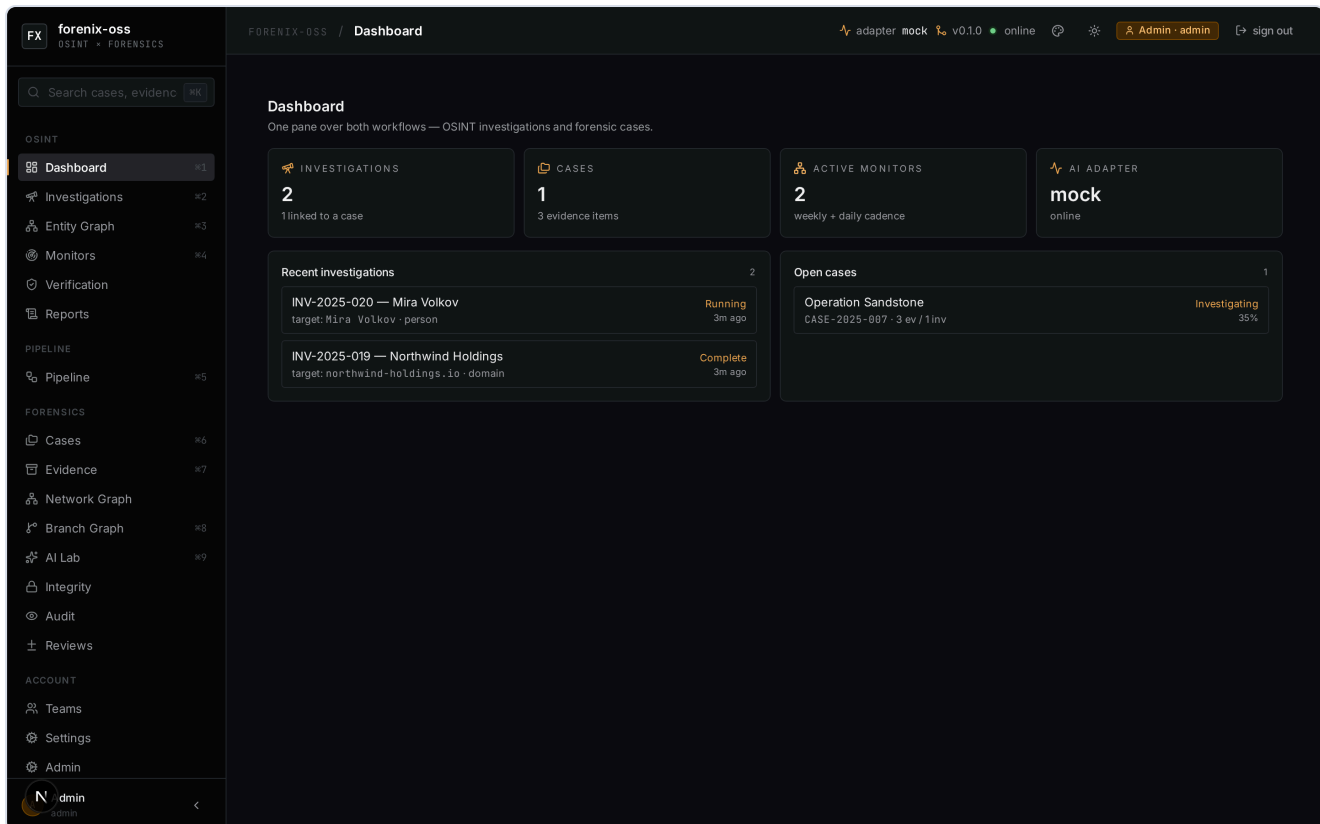
3.1 Install + seed

```
git clone https://github.com/thunderstornX/forenix-oss
cd forenix-oss
bun install
cp .env.example .env
bun run db:push
bun run db:seed
bun run dev
```

Open <http://localhost:3000>. (The screenshots in this manual were taken at port 3737 - the port is configurable.)

3.2 First screen

The landing view is the Dashboard. It pulls live counts from the database and renders both workflows on one pane.



What you're looking at:

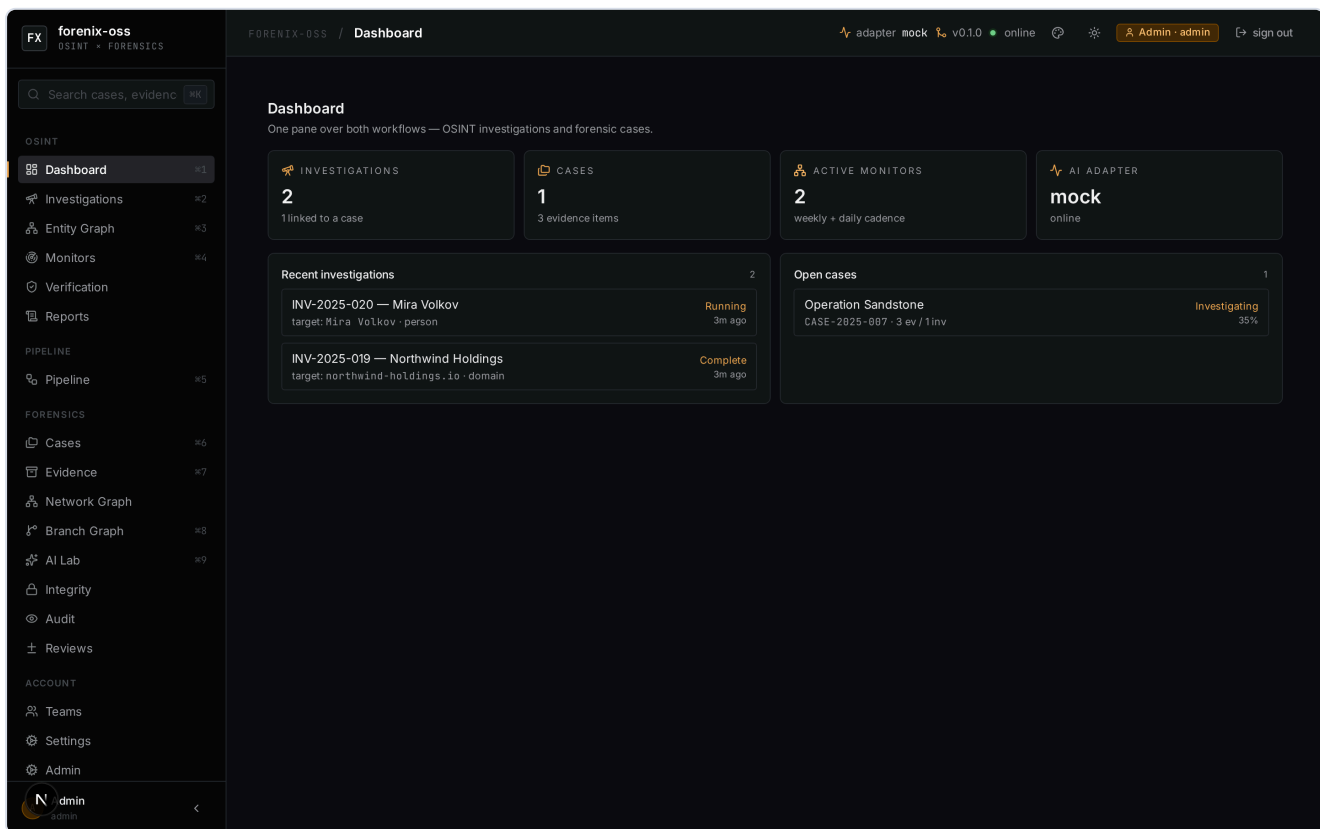
- **Sidebar (left).** Three sections: OSINT (Argus heritage), Pipeline (the bridge view), Forensics (ForenX heritage). The active view is highlighted in teal with a soft glow.
- **Top bar (centre).** The current view's name + the active AI adapter chip + the version + the online status.
- **Stat cards.** Investigations, Cases, Active monitors, AI adapter. Each is wired to a live query - refresh and the numbers update.
- **Recent investigations + Open cases.** Sortable preview rows pulled straight from the database.

Click any row in *Recent investigations* or *Open cases* to drill straight into the detail panel.

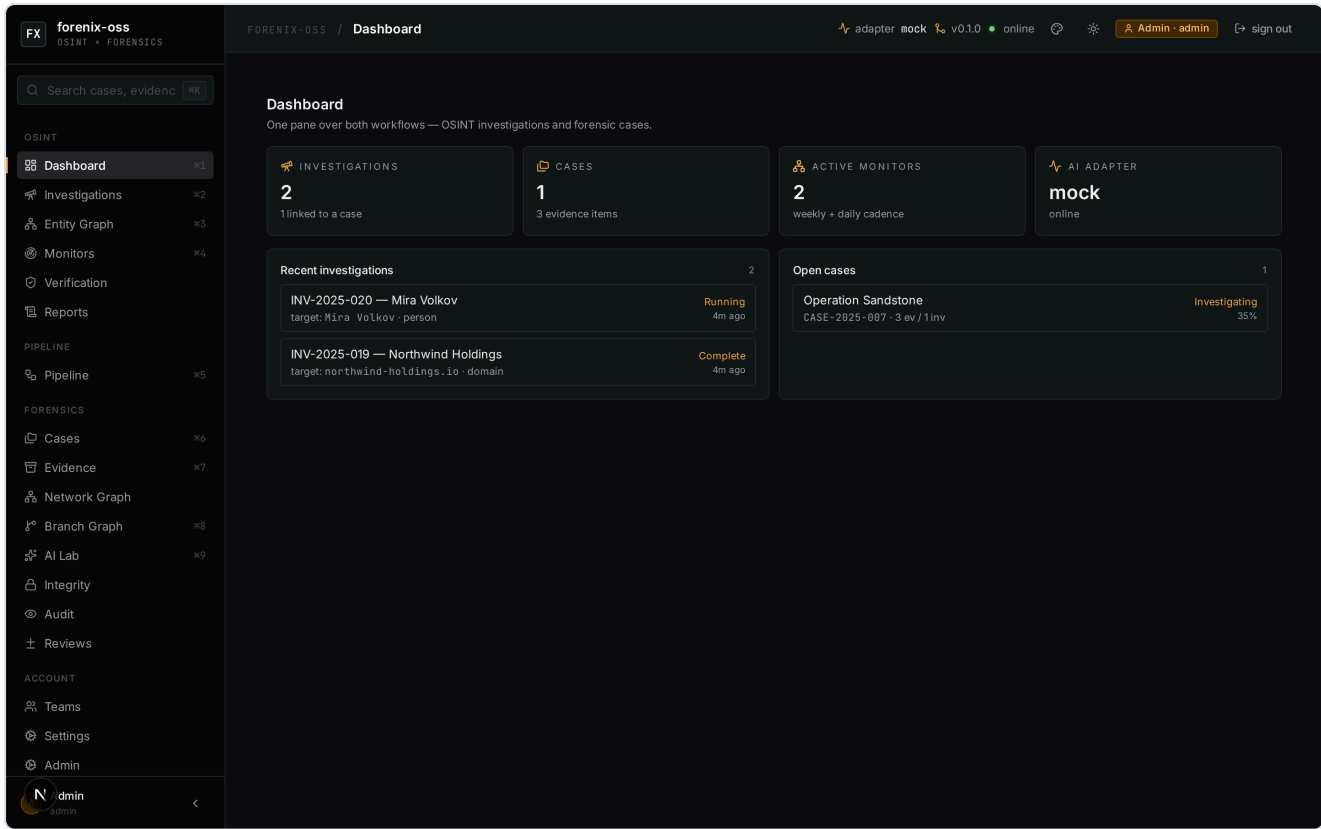
4. The workspace, in 60 seconds

The whole platform is one page (/) with a sidebar-driven view router. Switching views is **instant** - no route load, no re-fetch unless you ask.

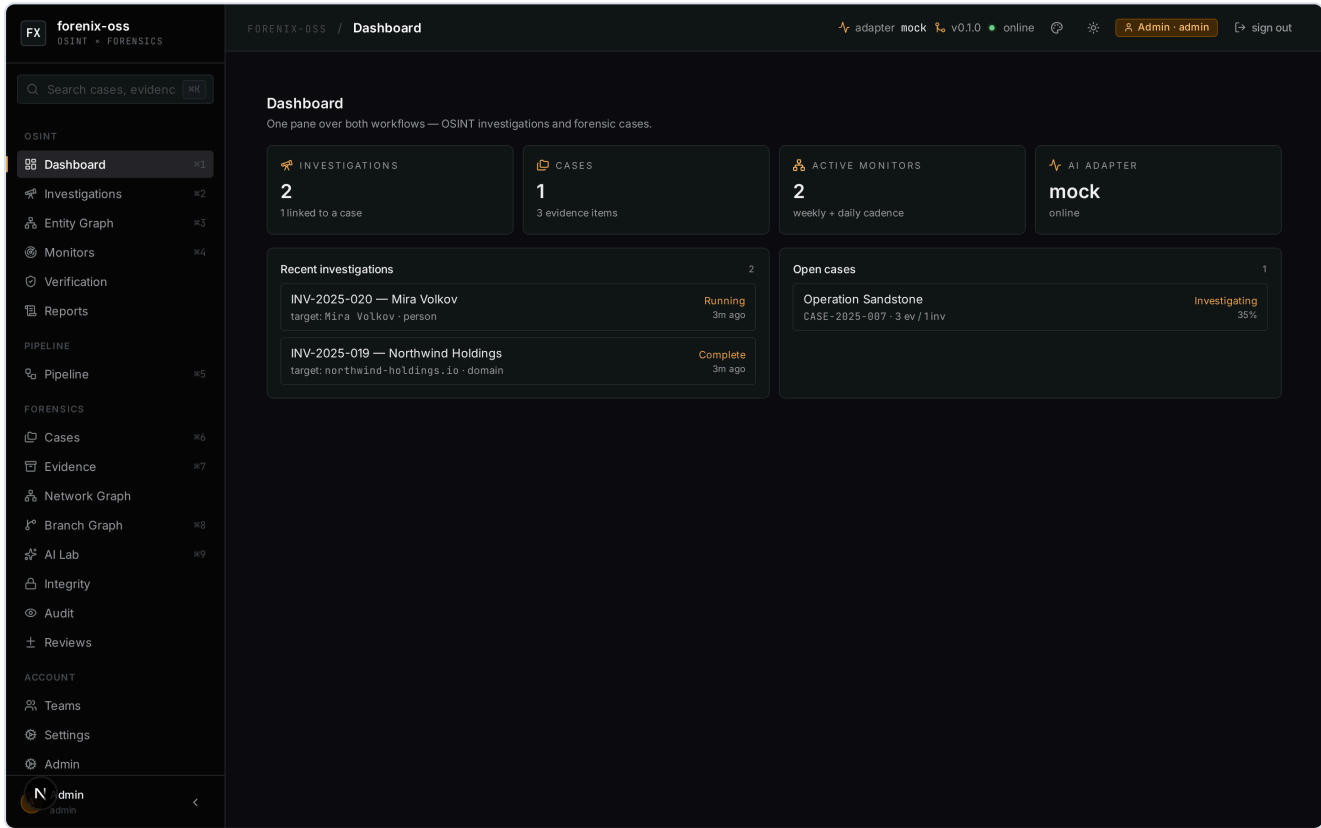
4.1 Sidebar



- **Brand** (top-left) - the platform name + the "OSINT x Forensics" tagline.
- **Search** button - opens the global command palette (⌘K).
- **OSINT section** - Dashboard, Investigations, Entity Graph, Monitors, Verification, Reports.
- **Pipeline section** - the bridge runner.
- **Forensics section** - Cases, Evidence, Network Graph, Branch Graph, AI Lab, Integrity, Audit, Reviews.
- **Collapse toggle** (bottom-left) - slim mode with just icons.



4.2 Top bar



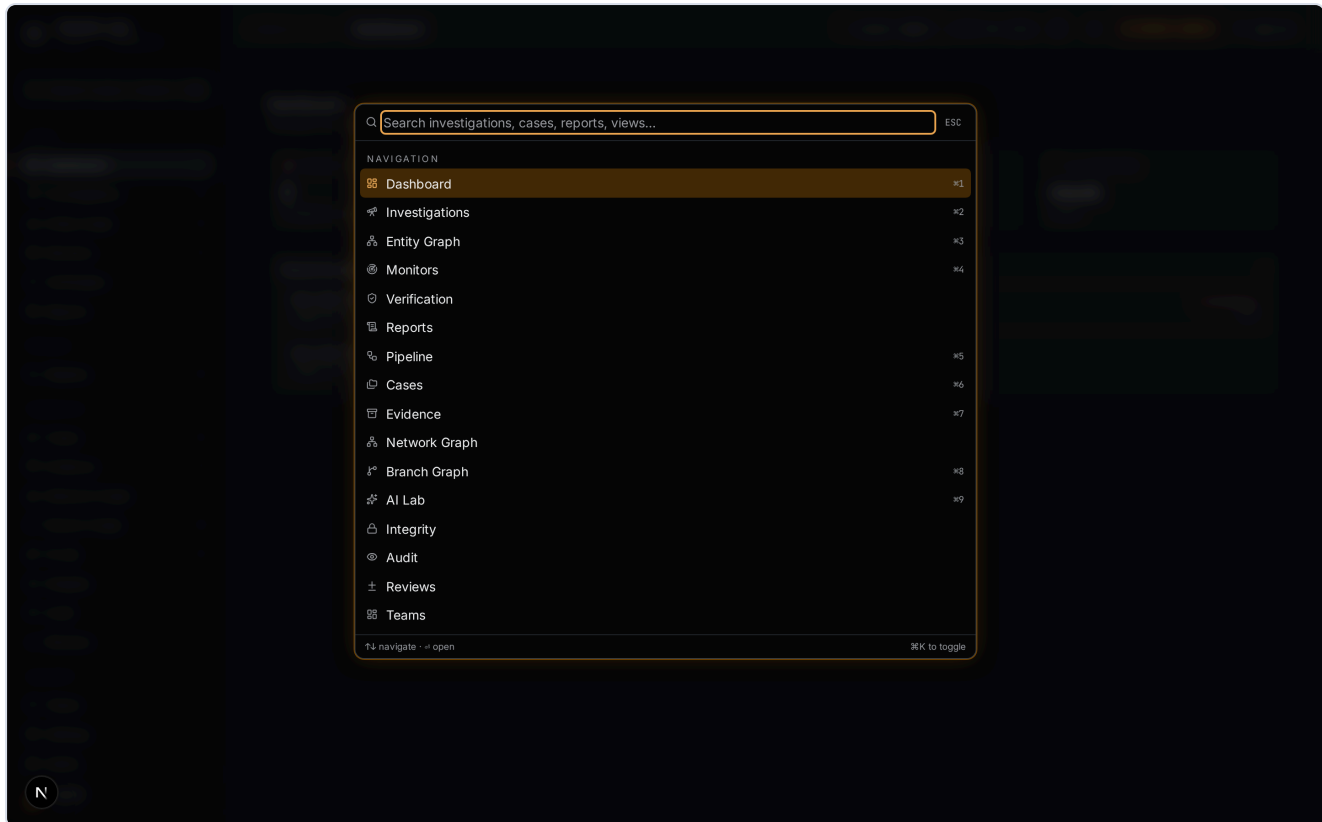
The top bar always shows:

- the **active view's title** (left);

- the **active AI adapter** (right) - useful when you've switched to a real model and want to verify the chip says `nvidia` or `openrouter` ;
- the **version** of the platform;
- the **online** status - green when the database + adapter are reachable.

4.3 Command palette (⌘K)

Press **⌘K** (or **Ctrl+K** on Linux/Windows) anywhere. The palette opens.



Type a few letters of any:

- nav item (e.g. `audit` , `entity`),
- investigation title,
- case number,
- report title.

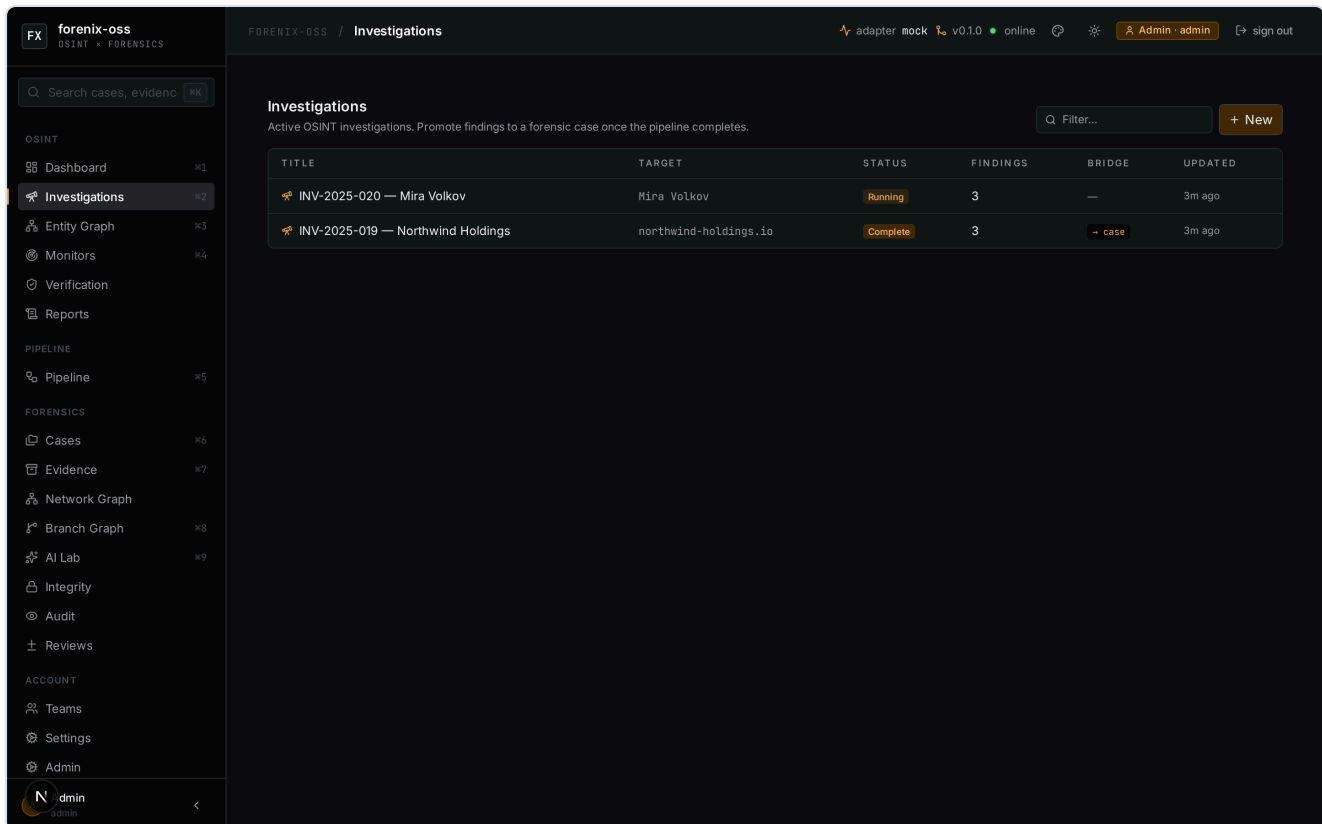
Use the arrow keys to navigate, Enter to open. Esc to close. This is the fastest way to move around the platform once your dataset gets non-trivial.

5. Workflow 1 - Your first OSINT investigation

This is the workflow you'll run dozens of times a week. We'll walk through it slowly the first time.

5.1 Start at Investigations

Click **Investigations** in the sidebar (or press **⌘2**).



The screenshot shows the 'forenix-oss' web application interface. The top navigation bar includes the 'forenix-oss' logo, 'OSINT' and 'FORENSICS' tabs, and a user profile 'Admin · admin' with a 'sign out' link. The sidebar on the left contains a search bar and a list of navigation items categorized under 'OSINT', 'PIPELINE', 'FORENSICS', and 'ACCOUNT'. The main content area is titled 'Investigations' and features a subtitle: 'Active OSINT investigations. Promote findings to a forensic case once the pipeline completes.' Below this is a table with columns: TITLE, TARGET, STATUS, FINDINGS, BRIDGE, and UPDATED. The table lists two investigations: 'INV-2025-020 — Mira Volkov' (Running) and 'INV-2025-019 — Northwind Holdings' (Complete). A '+ New' button is located in the top right corner of the table area.

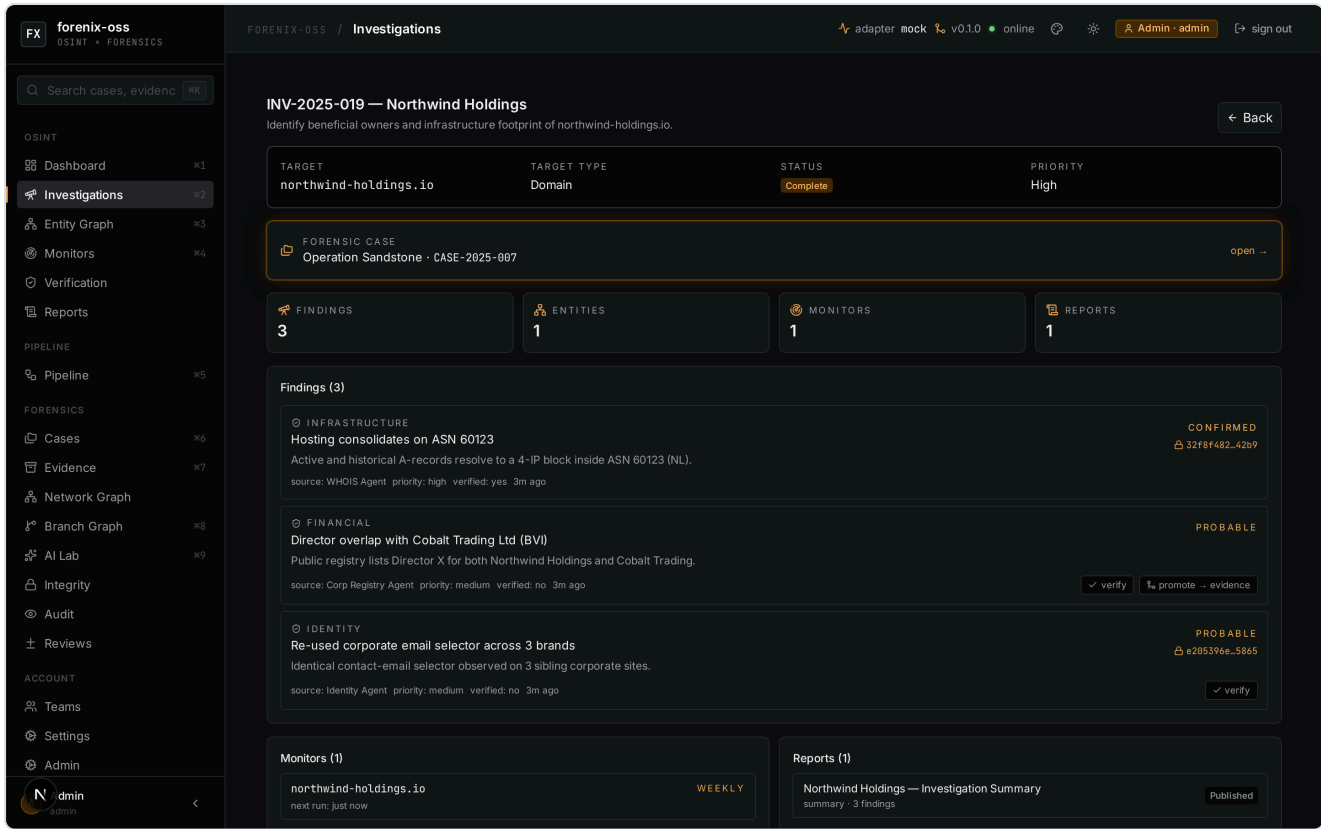
TITLE	TARGET	STATUS	FINDINGS	BRIDGE	UPDATED
INV-2025-020 — Mira Volkov	Mira Volkov	Running	3	—	3m ago
INV-2025-019 — Northwind Holdings	northwind-holdings.io	Complete	3	→ case	3m ago

What you see:

- A table of every investigation, newest first.
- Each row has its title, target, status, finding count, a bridge chip (if it's been linked to a forensic case), and the time it was last updated.
- A filter input above the table lets you type to narrow the list - it matches against title, target, target type, status, and priority.
- A **+ New** button opens the create-form.

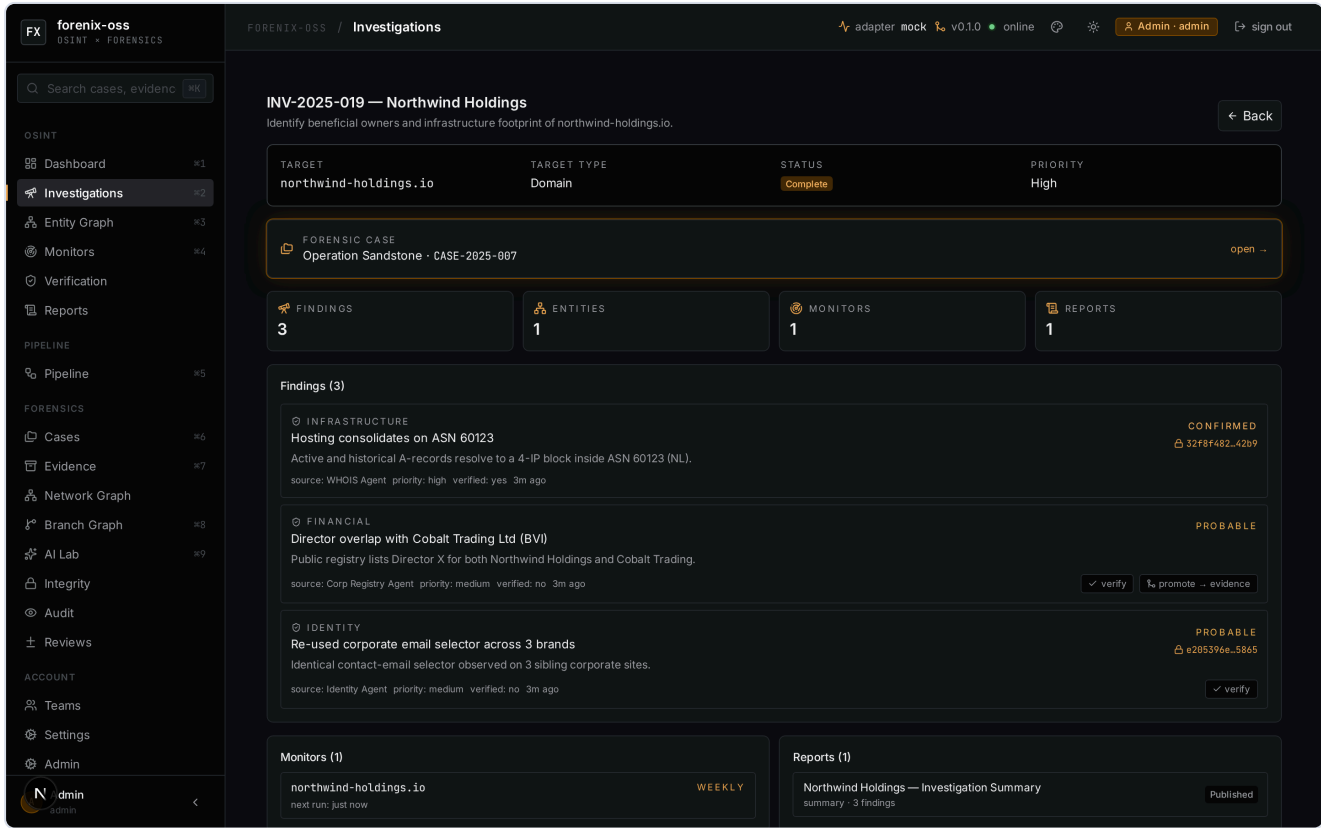
5.2 Open one to see the detail

Click any row - let's pick **INV-2025-019 - Northwind Holdings**.



What you're looking at:

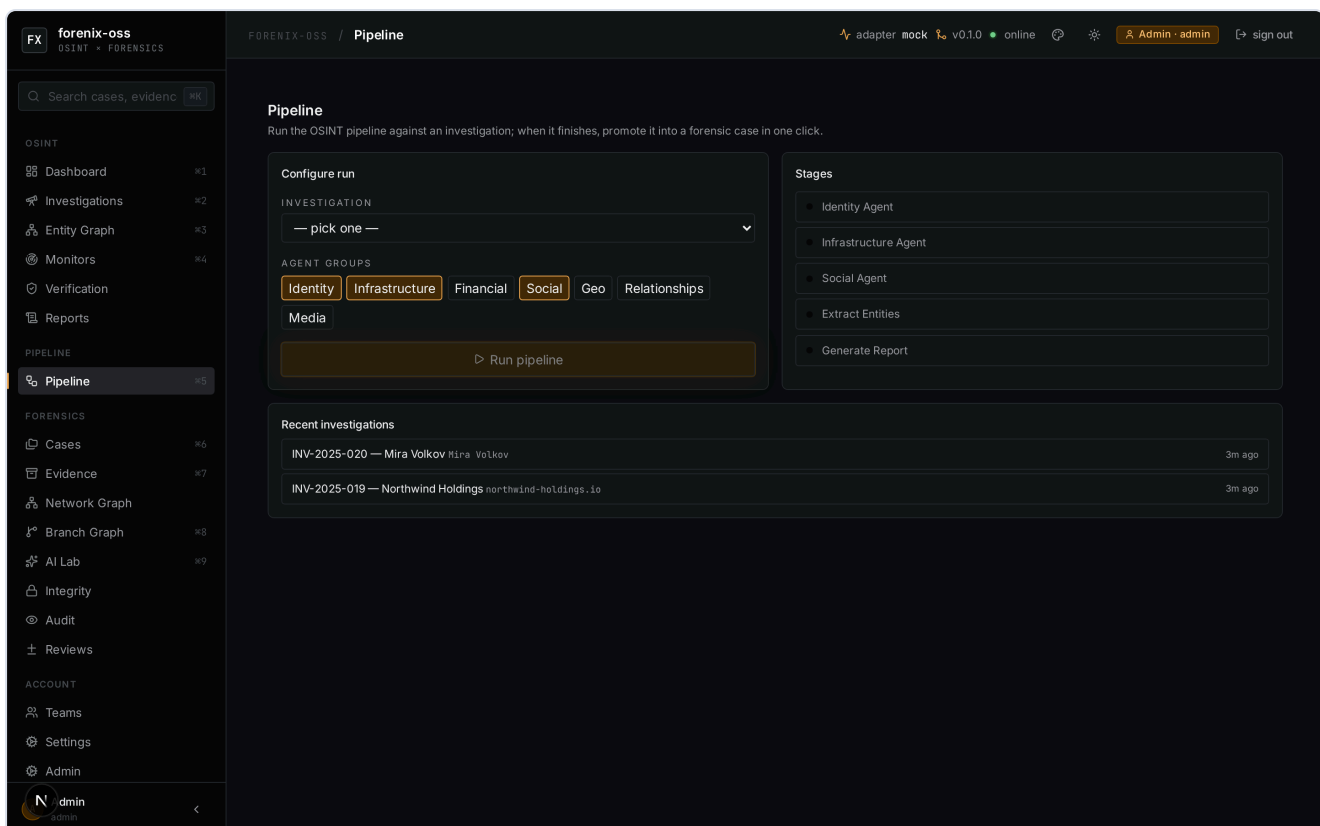
- **Header strip** - target, target type, status, priority. All the at-a-glance metadata.
- **Bridge card (the green one)**. This investigation is already linked to `CASE-2025-007 - Operation Sandstone` . Click it and the view switches straight into the case.



- **Stat row** - findings, entities, monitors, reports for this investigation.
- **Findings list** - every signal an agent produced, grouped by category. Each finding shows its confidence chip, priority, source, the reasoning trace, and (if it's been promoted) the evidence hash.
- **Action buttons on each finding row:**
 - **verify** - flips the finding's confidence from `unverified` to `confirmed` and records you as the verifier. Writes a `verify_finding` audit row.
 - **promote -> evidence** - only visible if the investigation is bridged to a case AND the finding hasn't been promoted yet. Creates an Evidence row with its own initial commit and stamps `Finding.evidenceId`.
- **Monitors / Reports** - two side-by-side cards listing the monitor cadences and the published reports.
- **Audit trail** (bottom) - every state change for this investigation, in chain order.

5.3 Run the pipeline against a fresh target

Switch to the **Pipeline** view (sidebar, or ⌘5).

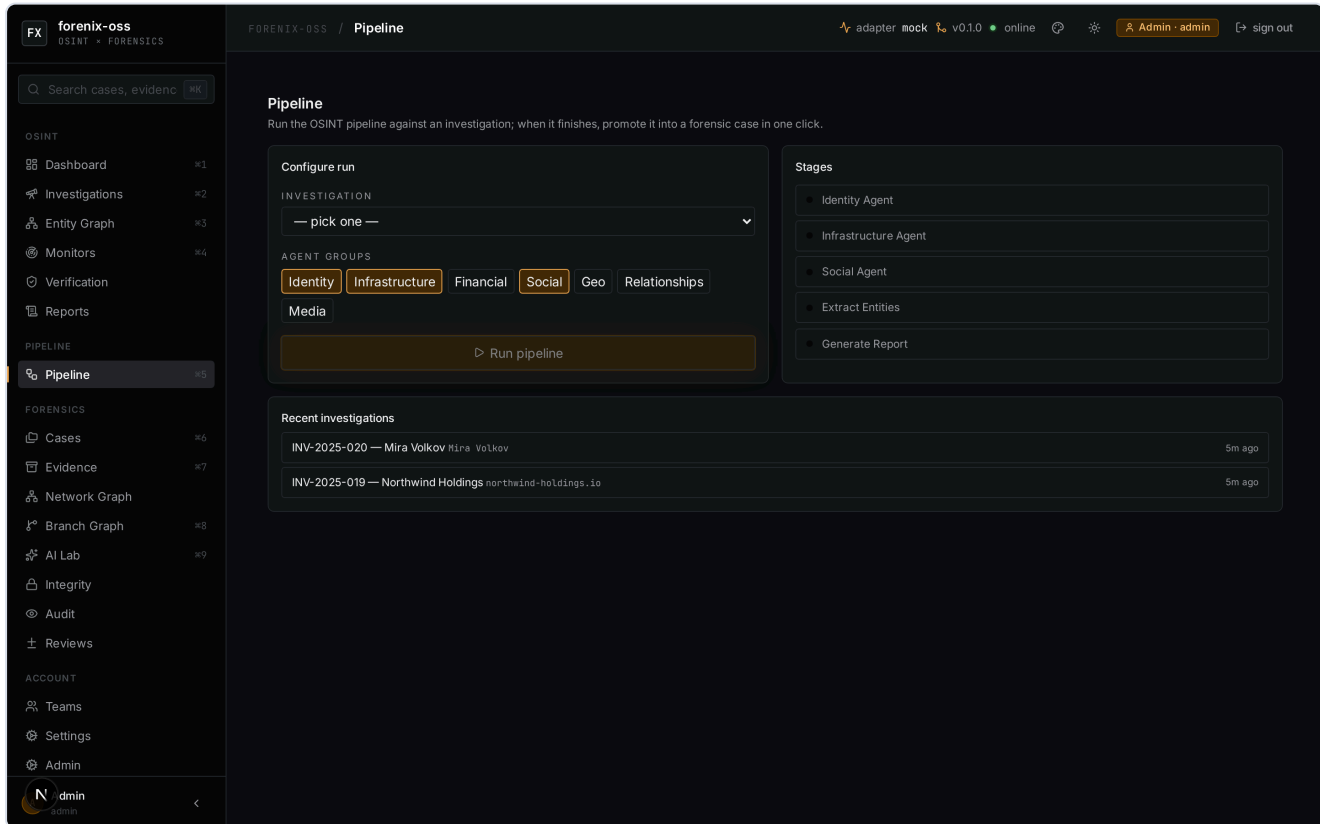


The Pipeline runner is the headline feature. It takes an existing investigation, fans out the AI agents in parallel against the target, consolidates the findings, extracts entities, and generates a draft report - all auditable, all in one button.

To run:

1. Pick an investigation from the dropdown (or click one in *Recent investigations* below - clicking sets the selection).
2. Toggle the agent groups you want to run. The defaults are `identity`, `infrastructure`, `social` - three covers most targets.

- (Optional) override the AI adapter by passing `?adapter=nvidia` in the URL or `adapter:"nvidia"` in the body if you're calling the API directly. By default the active adapter is the one in `.env`.
- Click **Run pipeline**.



While the pipeline runs:

- Each stage in the right-hand panel transitions `idle -> running -> done`.
- A `Loader2` spinner shows what's currently in flight.

When the run completes (typically 30-90 seconds on hosted LLMs), the *Pipeline complete* card appears with the counts:

- findings** - how many discrete signals were produced.
- entities** - how many people / orgs / domains / IPs were extracted from those findings.
- relations** - how many directed connections between entities were inferred.
- report** - a draft markdown report tied to the investigation.

The next step - and this is what makes forenix-oss different - is the **Open forensic case ->** button.

6. Workflow 2 - Bridge findings to a forensic case

This is the workflow that turns OSINT into evidence.

6.1 What the bridge does

Clicking **Open forensic case ->** does five things in one transaction:

1. **Mints a Case** with an auto-generated `CASE-YYYY-NNN` number, status `open` , priority inherited from the investigation.
2. **Creates a `main` branch** on that case (forensic cases are structured like a Git repo).
3. **Sets `Investigation.caseId`** - the link that powers the green bridge chip you saw earlier.
4. **Promotes every finding** to an Evidence row, computes a SHA-256 over the finding's content, and adds an initial `add:` commit to the case's `main` branch.
5. **Stamps `Finding.evidenceId`** so each finding now points at the evidence row it became.

Every one of those five steps writes an audit row. The chain stays valid throughout.

6.2 The bridged case detail

After the bridge, you land on the new case. It looks like this:

The screenshot shows the 'forenix-oss' web application. The top navigation bar includes 'FORENIX-OSS / Cases' and user information 'Admin · admin'. The sidebar on the left lists various sections: OSINT (Dashboard, Investigations, Entity Graph, Monitors, Verification, Reports), PIPELINE (Pipeline), FORENSICS (Cases, Evidence, Network Graph, Branch Graph, AI Lab, Integrity, Audit, Reviews), and ACCOUNT (Teams, Settings, Admin). The main content area is titled 'Operation Sandstone' and describes a cross-jurisdictional shell-corporation network. It features a case header with details like CASE #, STATUS (Investigating), PRIORITY (High), and PROGRESS (35%). Below this are cards for 'Linked investigations (1)', 'EVIDENCE 3', 'BRANCHES 2', 'MERGE REQUESTS 0', and 'ASSIGNEES 2'. The 'Evidence (3)' section lists items like 'pcap-handshake-fragment.pcap' and 'whois-snapshot-northwind.json', each with a commit chain and a 'seal' button.

Top to bottom:

- **Case header** - number, status, priority, progress bar.
- **Linked investigations card** - every investigation that feeds this case. Click any row to jump back to its investigation detail.
- **Stat row** - evidence, branches, merge requests, assignees.
- **Evidence list** - and this is the heart of the page. Each evidence row shows:
 - name, type, MIME, description,
 - **truncated SHA-256 hash** in monospace,
 - **status chip** - `collected` / `verified` / `sealed` ,
 - a **Seal** button (we'll get to it),

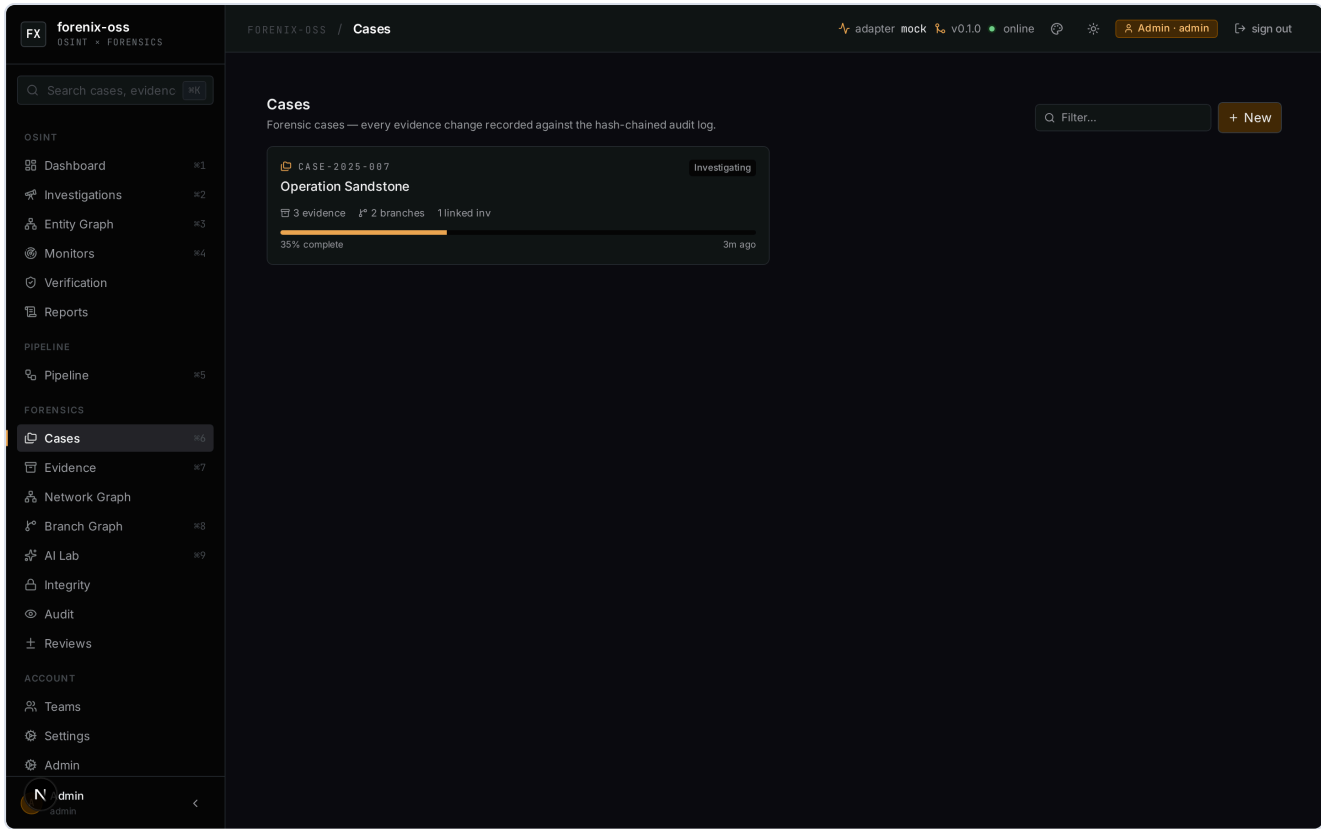
- and the **full commit chain inline**.

The screenshot displays the 'forenix-oss' web application interface. The left sidebar contains navigation links for OSINT, PIPELINE, FORENSICS, and ACCOUNT. The main content area shows the 'Cases' view for 'Operation Sandstone'. At the top, there's a header with 'FORENIX-OSS / Cases' and user information 'Admin · admin'. Below this, a summary card for 'CASE-2025-007' shows its status as 'Investigating', priority as 'High', and progress as '35%'. A section titled 'Linked investigations (1)' lists a domain 'INV-2025-019 — Northwind Holdings' with a 'Complete' status. Below this, four summary boxes show 'EVIDENCE 3', 'BRANCHES 2', 'MERGE REQUESTS 0', and 'ASSIGNEES 2'. The 'Evidence (3)' section displays two items: a 'pcap-handshake-fragment.pcap' file and a 'whois-snapshot-northwind.json' file. Each item includes a 'COMMIT CHAIN (2)' with details on additions and verifications, along with a 'seal' icon. The bottom of the interface shows a 'DOCUMENT · MESSAGE/RFC822' entry.

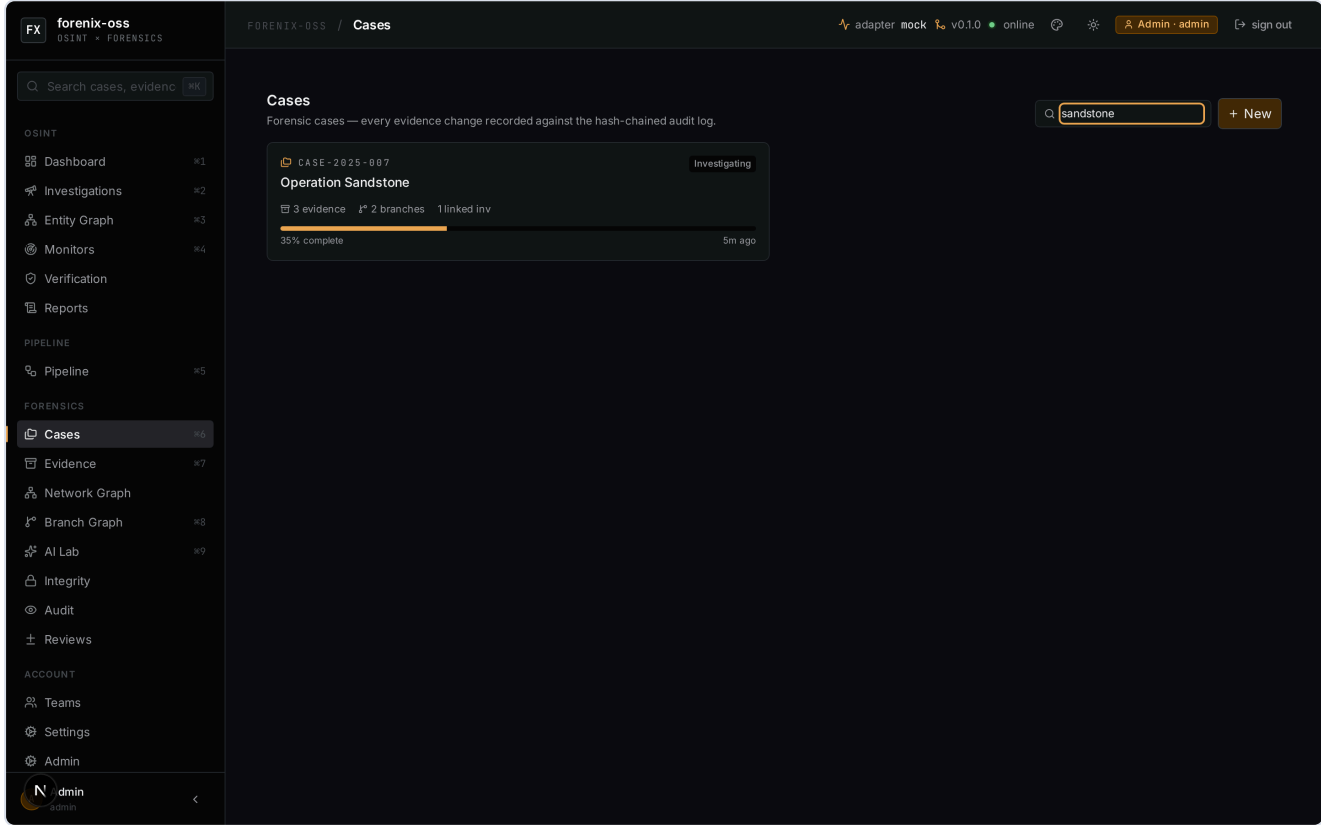
- **Branches sidebar** - every branch on this case (`main` , `evidence-review` , etc.), each colour-coded.
- **Assignees + agents** - humans and AI agents working on this case.
- **Recent audit entries** - the last 10 audit-chain rows scoped to this case.

6.3 The Cases list

If you'd rather start from the Cases view, click **Cases** (⌘6).



Cards show case number, title, status, progress bar, and high-level counts. Click any card to drill in. The inline filter matches across `title`, `caseNumber`, `status`, and `priority`.

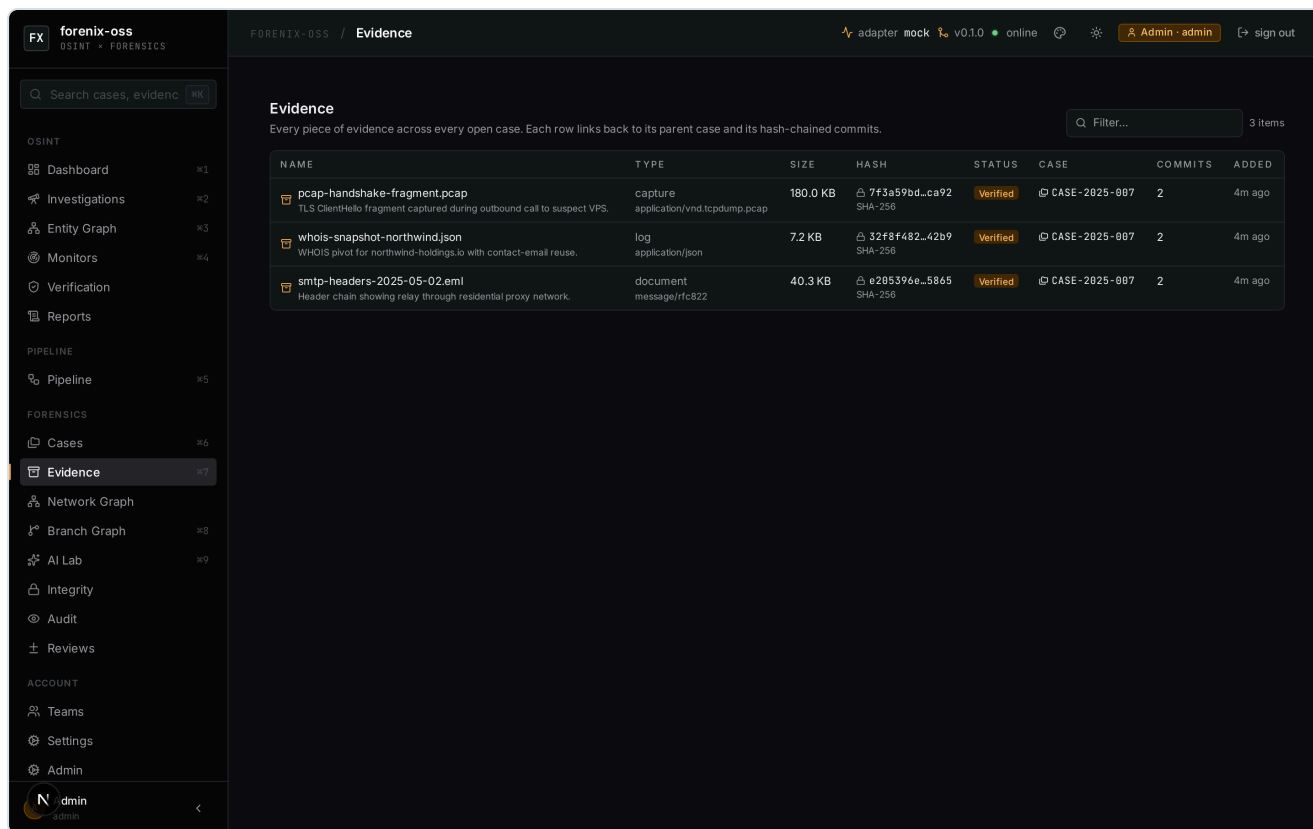


7. Workflow 3 - Evidence + chain of custody

Evidence is the unit of forensic work. Each Evidence row carries a content hash + a commit history. Forenix-oss is one of the very few platforms where every change to a piece of evidence shows up on a tamper-evident chain.

7.1 The Evidence inventory

Switch to **Evidence** (§7).



The screenshot shows the forenix-oss web interface. The left sidebar contains navigation links for OSINT, PIPELINE, and FORENSICS. The main area displays the 'Evidence' view with a table of evidence items. The table has columns: NAME, TYPE, SIZE, HASH, STATUS, CASE, COMMITS, and ADDED. Three items are listed: 'pcap-handshake-fragment.pcap', 'whois-snapshot-northwind.json', and 'smtp-headers-2025-05-02.eml'. Each item has a 'Verified' status and is linked to 'CASE-2025-007'.

NAME	TYPE	SIZE	HASH	STATUS	CASE	COMMITS	ADDED
pcap-handshake-fragment.pcap TLS ClientHello fragment captured during outbound call to suspect VPS.	capture application/vnd.tcpdump.pcap	180.0 KB	7f3a59bd...ca92 SHA-256	Verified	CASE-2025-007	2	4m ago
whois-snapshot-northwind.json WHOIS pivot for northwind-holdings.io with contact-email reuse.	log application/json	7.2 KB	32f8f482...42b9 SHA-256	Verified	CASE-2025-007	2	4m ago
smtp-headers-2025-05-02.eml Header chain showing relay through residential proxy network.	document message/rfc822	40.3 KB	e285396e...5865 SHA-256	Verified	CASE-2025-007	2	4m ago

The Evidence view shows every piece of evidence across every case. Useful when you want to find something by hash, or inventory the state of an entire investigation.

Columns: name, type/MIME, size, **hash** (truncated), status, parent case, commit count, added-at. Filter works across all of these.

7.2 Seal an evidence row

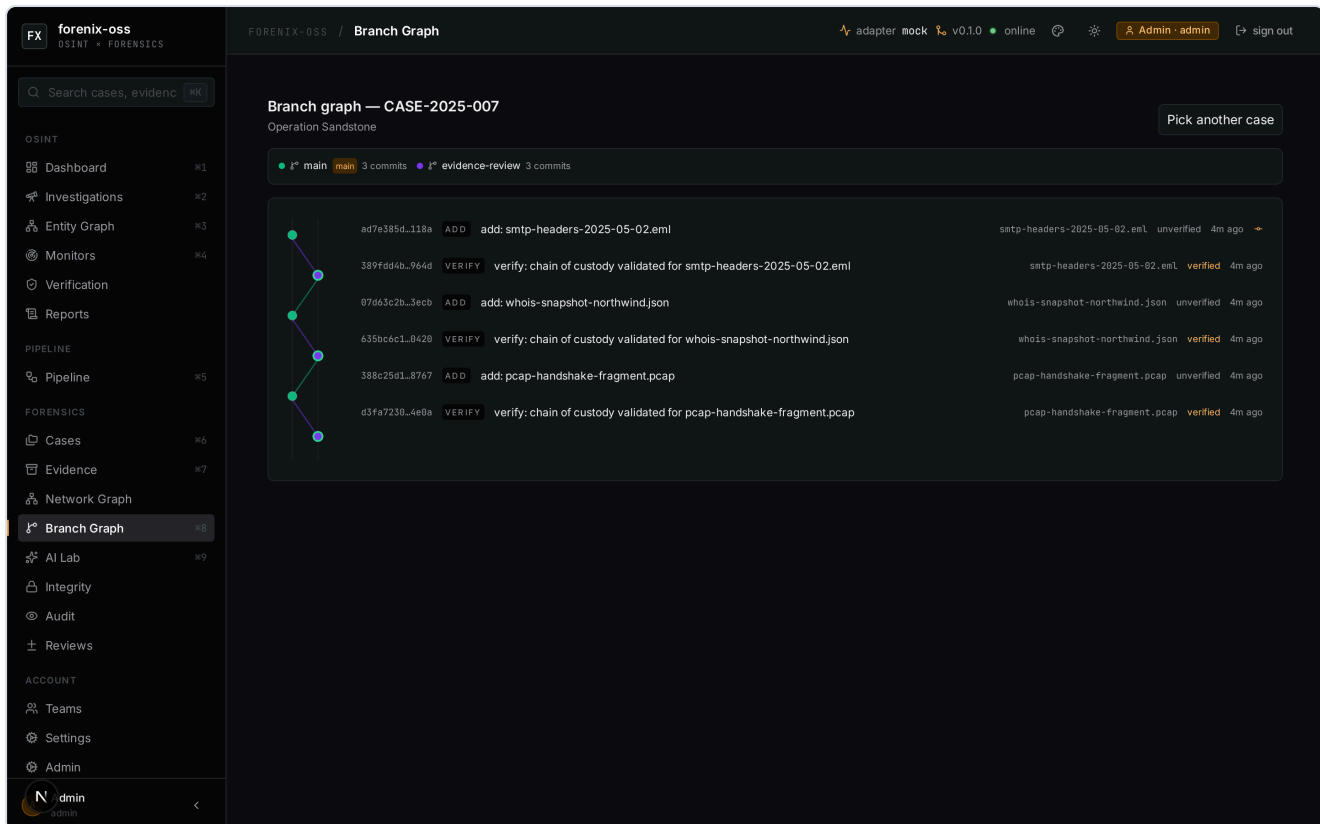
A **sealed** evidence row is immutable - no further commits are accepted, and the row's hash is locked into the chain.

To seal, click the **Seal** button on the right side of any evidence card on the case-detail page. The status chip flips to **sealed**, a **seal** commit is written to **main**, and an audit row is added.

This is the action you take when the case team has signed off and the evidence is "ready for court."

7.3 The Git-style branch graph

For visual reasoning about the commit history, switch to **Branch Graph** (§8) and pick the case you want to inspect.



What you see:

- One **swimlane per branch**, leftmost is `main`.
- One **dot per commit**, coloured to match the branch.
- Verified commits get a **green ring** around the dot.
- The right-hand panel lists each commit's hash, change type (`add` / `verify` / `seal` / `merge`), evidence it touched, verified state, and time.

This is the page you'd put in front of a sceptical reviewer.

8. Workflow 4 - Verify the audit chain

The audit chain is what makes forenix-oss claim "court-grade custody." It's a SHA-256 forward chain over every state-changing write.

8.1 Open the Audit view

Click **Audit** in the sidebar.

Audit log

Append-only chain across both workflows. Every write computes sha256(prevHash | action | entity | entityid | iso(t)).

Q: action, entity, hash... 9 entries

HASH	ACTION	ENTITY	SCOPE	WHEN
4b465bca...7f26 -- 00000000...0000	create_user	User cmp97rsz8000	—	5m ago
bb10de21...a104 -- 4b465bca...7f26	create_user	User cmp97rt01000	—	5m ago
1b117e08...6856 -- bb10de21...a104	create_user	User cmp97rt05000	—	4m ago
61830660...082e -- 1b117e08...6856	create_investigation	Investigation cmp97rt2c001	inv	4m ago
52ad3c30...d86b -- 61830660...082e	run_pipeline	Investigation cmp97rt2c001	inv	4m ago
04361111...7fc1 -- 52ad3c30...d86b	create_case	Case cmp97rt0f000	case	4m ago
7003e873...d45b -- 04361111...7fc1	promote_to_evidence	Finding f-1	case	4m ago
7c363f93...3a5d -- 7003e873...d45b	verify_evidence	EvidenceCommit cmp97rt0f000	case	4m ago
bd73298b...44ab -- 7c363f93...3a5d	generate_report	Report r-1	inv	4m ago

The table shows every audit row in chain order. Per row:

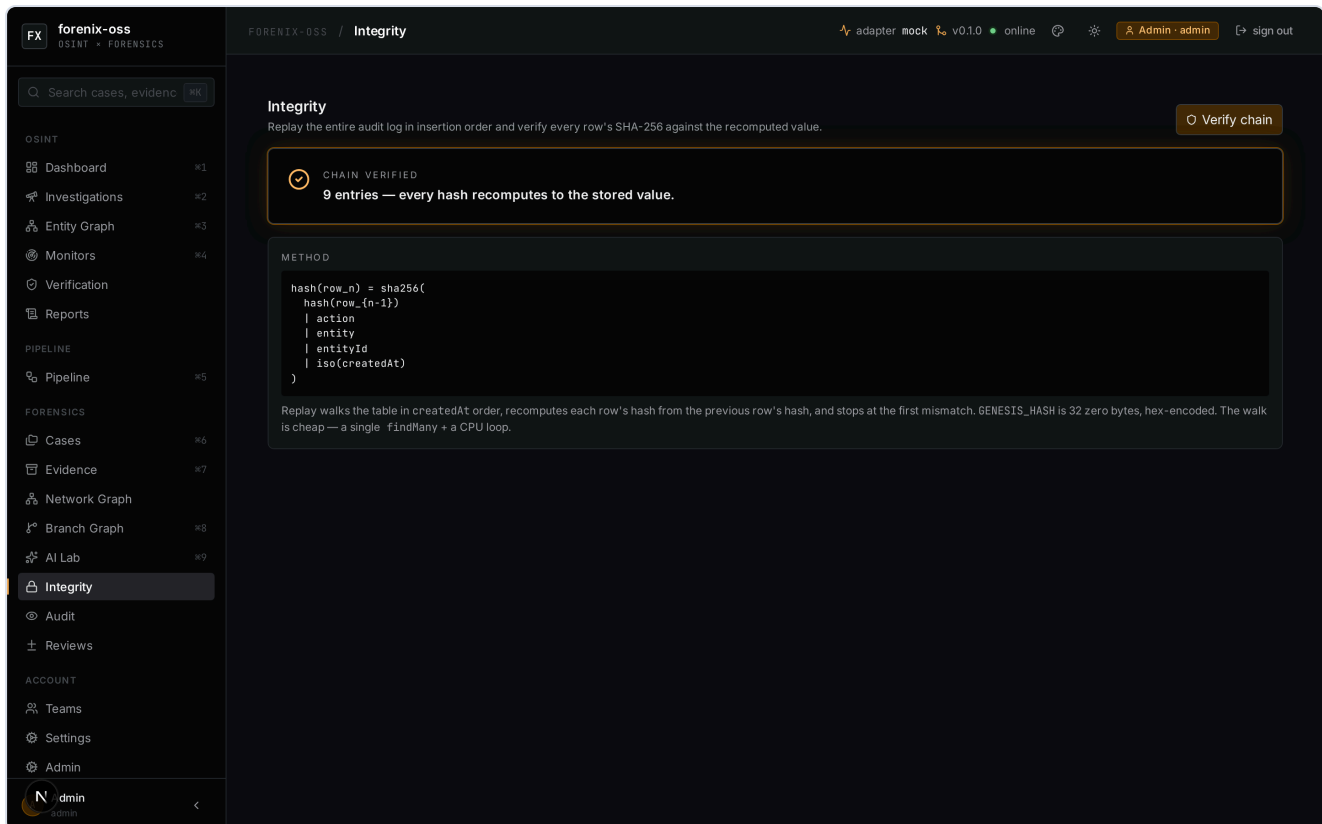
- the SHA-256 hash of the row itself (monospace, truncated for display);
- the **previous row's hash** inline directly below - visual proof of the chain;
- the action (`pipeline_started` , `agent_identity_completed` , `promote_finding_to_evidence` , `seal_evidence` , ...);
- the entity it acted on (Investigation / Case / Finding / ...);
- the scope - clickable jump links to the investigation and/or case;
- the relative time.

The filter input matches on action, entity, hash. Type `seal` to see every sealing event; type `pipeline` to see every pipeline event; etc.

A row whose `prevHash` doesn't equal the previous row's `hash` gets a red tint automatically.

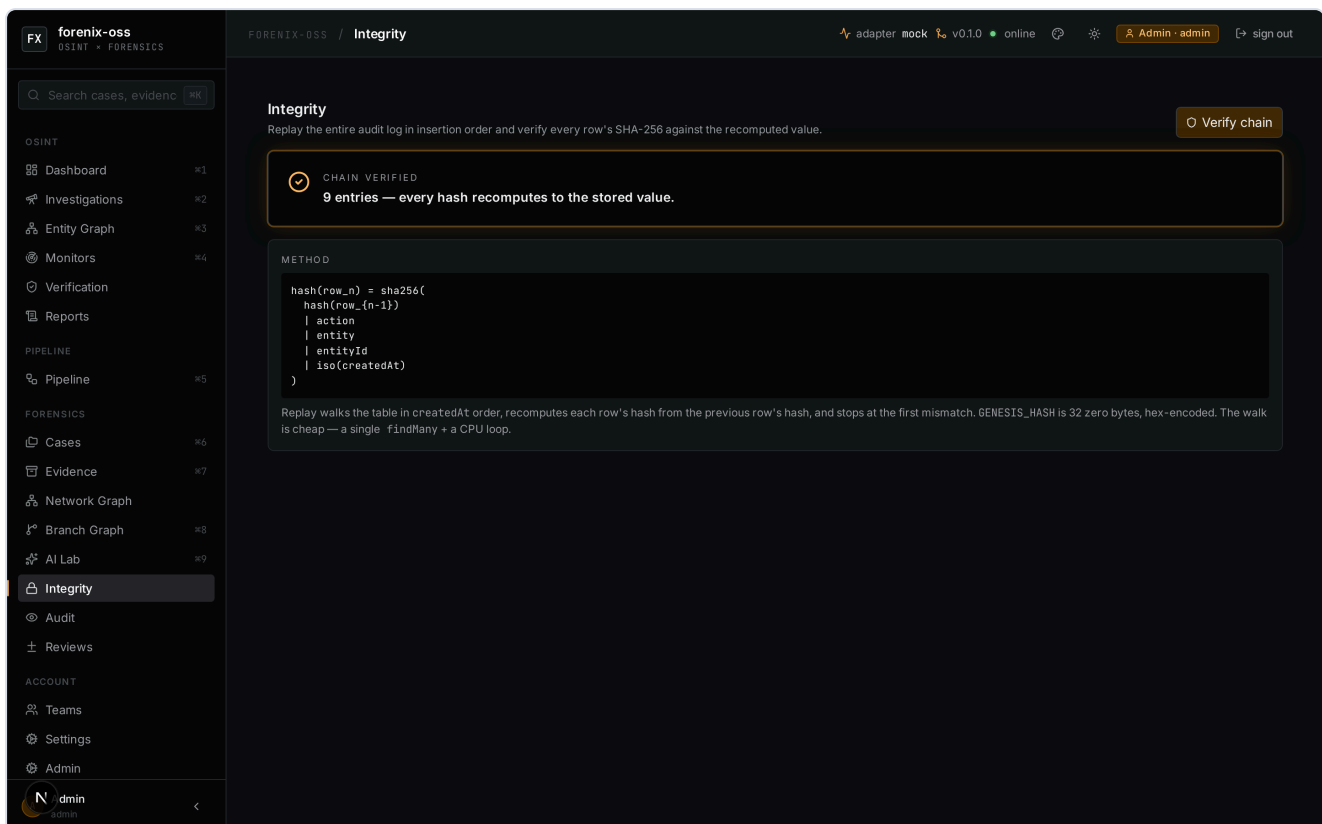
8.2 Verify the chain end-to-end

Now click **Integrity**.



The page shows the method (the SHA-256 input ordering), but no verdict yet. Click **Verify chain**.

The button calls `GET /api/audit/verify` which walks the entire table in `createdAt` order, recomputes every row's hash from the previous row's `hash`, and reports the result.



The green panel above is the real screenshot from a verified chain on the demo dataset - *21 entries, every hash recomputes to the stored value.*

If verification fails, the panel turns red and shows the row id where the chain broke, the expected hash, and the stored hash - all the information a forensic investigator needs to triage the event.

8.3 Verifying the chain offline

Anyone with read access to the database can verify the chain without our cooperation. From `docs/07-SECURITY.md` :

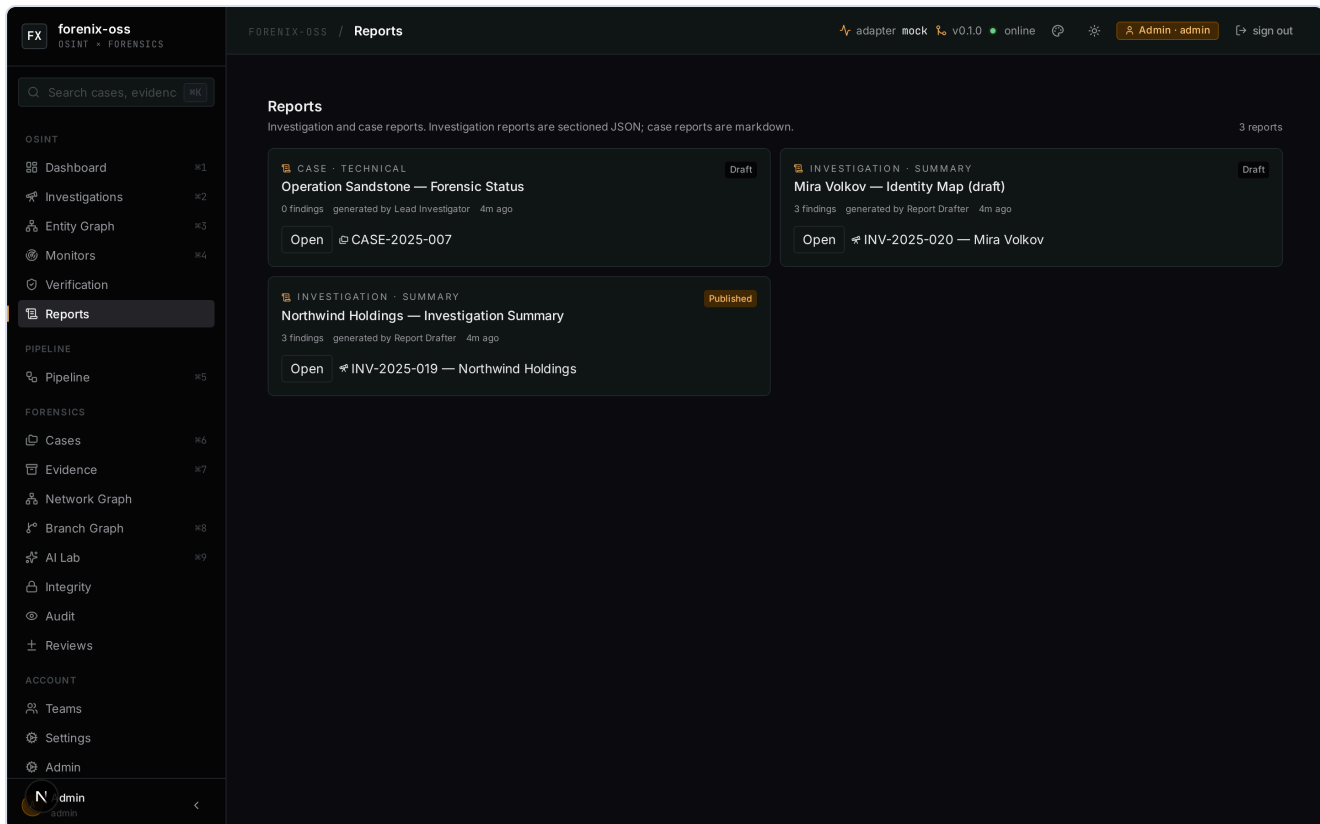
```
import csv, hashlib
GENESIS = "0" * 64
prev = GENESIS
for r in csv.DictReader(open("audit.csv")):
    h = hashlib.sha256("|".join([
        prev, r["action"], r["entity"], r["entityId"] or "",
        r["createdAt"]
    ]).encode("utf-8")).hexdigest()
    assert r["prevHash"] == prev and r["hash"] == h, f"BROKEN at {r['id']}"
    prev = r["hash"]
print("chain OK")
```

12 lines of Python. That's the entire trust anchor.

9. Workflow 5 - Generate and read reports

9.1 The Reports list

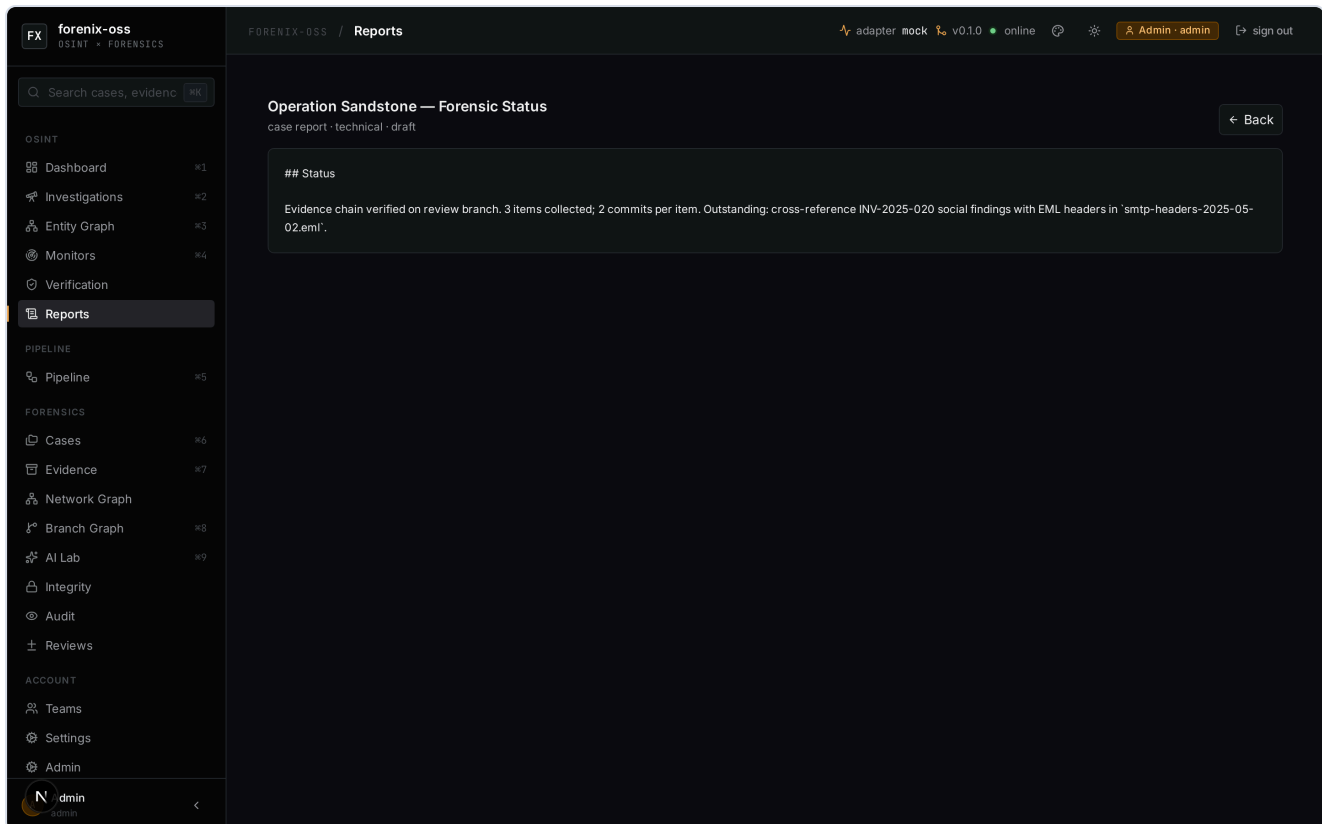
Click **Reports** in the sidebar.



The list shows every report - both *investigation* reports (sectioned JSON, the Argus shape) and *case* reports (markdown, the ForenX shape). Each card shows source, status, finding count, generator, and timestamps. Click **Open** to read.

9.2 Reading a report

The detail view auto-detects whether the report has `sections` (JSON) or `content` (markdown) - or both - and renders each accordingly.



For an analyst, this is the closest thing to a "deliverable" - copy/paste the section text into your write-up, or export the markdown directly.

9.3 How a report gets generated

Reports are produced automatically as the *last stage* of the Pipeline runner. The active AI adapter is called with `generateReport(investigation, findings)` and asked to produce a markdown report in this shape:

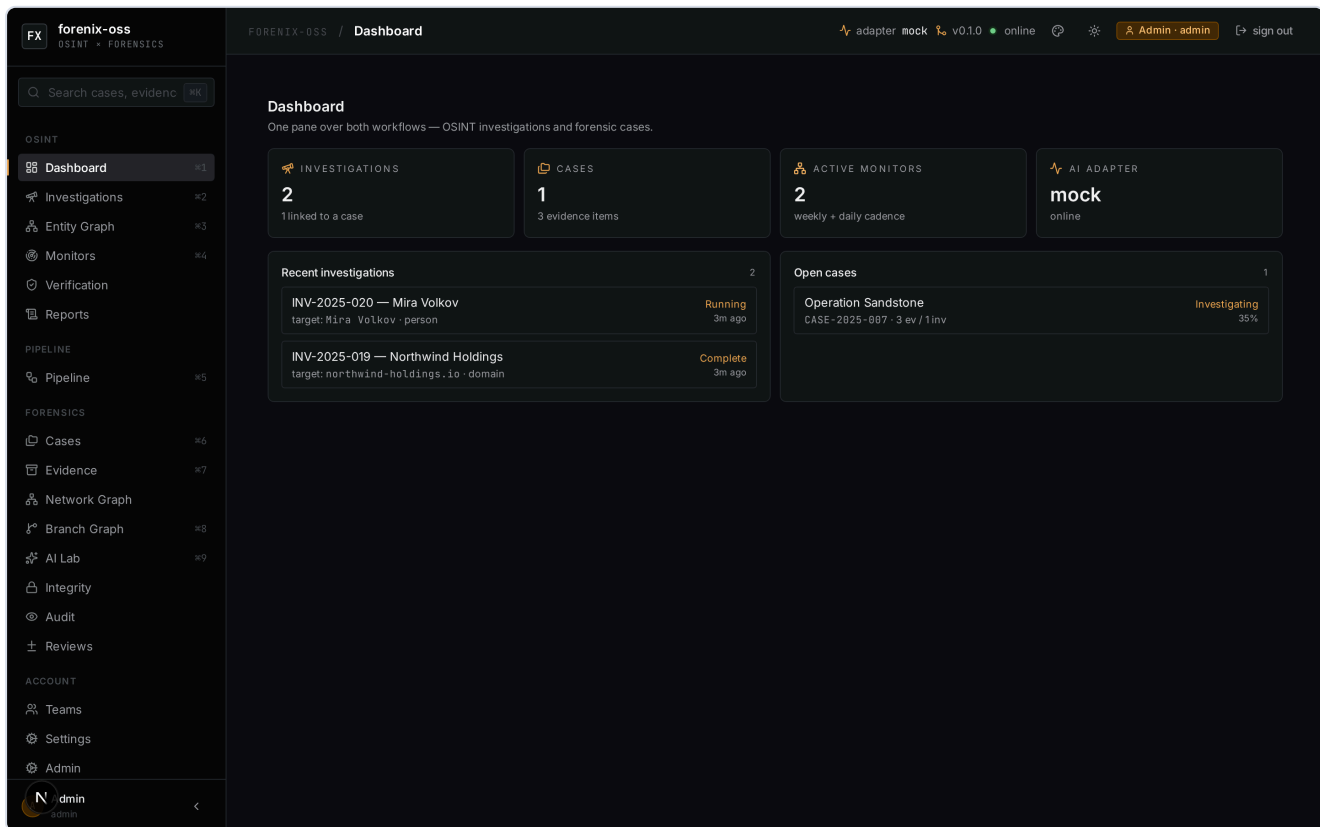
- `# Executive Summary` - 3-5 sentences.
- `## Findings` - grouped by agent group.
- `## Entity overview` - names mentioned in the findings.
- `## Recommended next steps` - 3-5 bullet points.

The report is stored on the `Report` table with `source = "investigation"` and gets attached to the parent investigation. You don't need to do anything - just run the pipeline.

10. Feature reference - every view

This section is a sweep across every view. For each one we explain *what it is*, *what problem it solves*, and *what it deliberately does not claim to do*.

10.1 Dashboard



- **What:** four stat cards + a *Recent investigations* preview and an *Open cases* preview.
- **Solves:** the analyst's "where do I start today?" problem.
- **Does not:** show charts/trendlines. No per-user scope (you see everything you have access to).

10.2 Investigations

The screenshot shows the 'Investigations' page in the forenix-oss application. The sidebar on the left contains navigation links for OSINT (Dashboard, Investigations, Entity Graph, Monitors, Verification, Reports), PIPELINE (Pipeline), and FORENSICS (Cases, Evidence, Network Graph, Branch Graph, AI Lab, Integrity, Audit, Reviews). The main content area is titled 'Investigations' and includes a subtitle 'Active OSINT investigations. Promote findings to a forensic case once the pipeline completes.' Below this is a table with the following data:

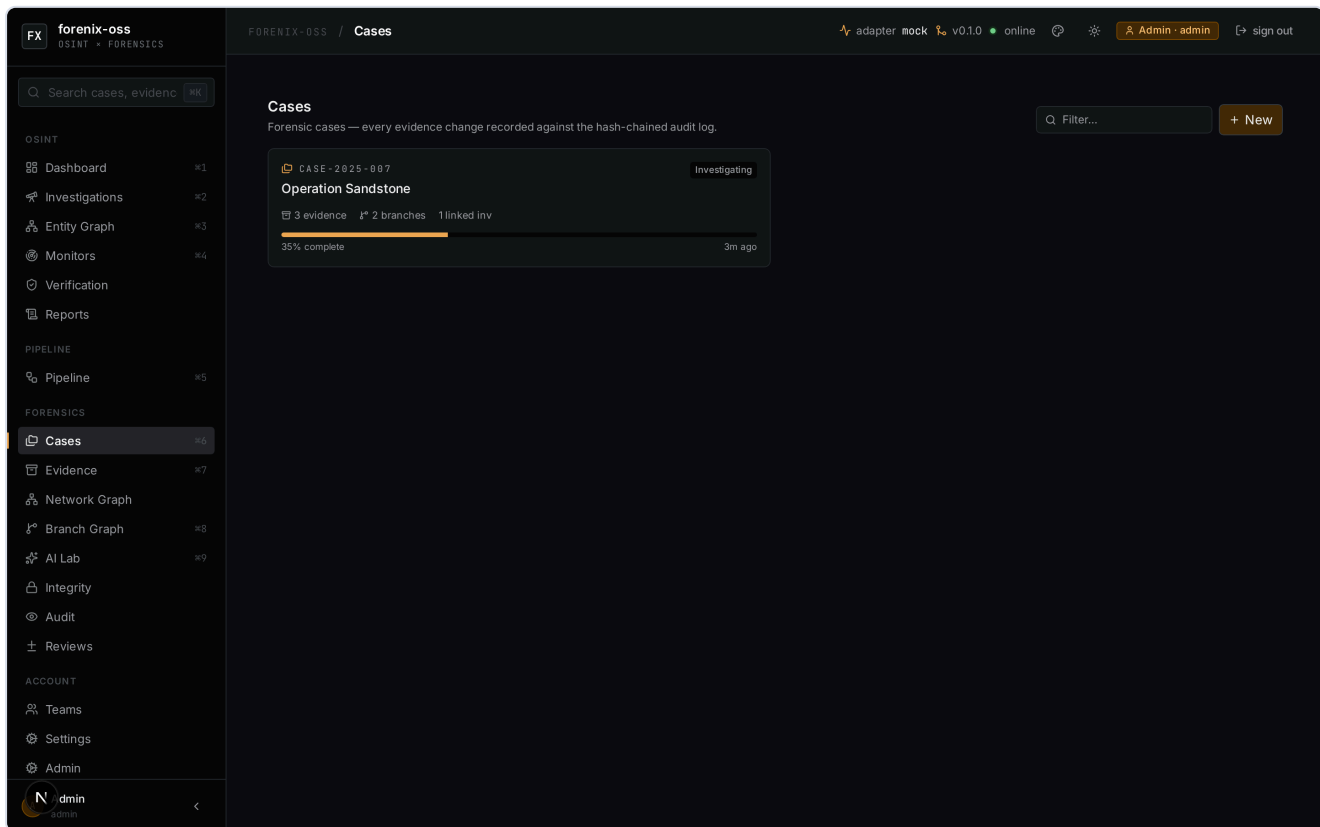
TITLE	TARGET	STATUS	FINDINGS	BRIDGE	UPDATED
INV-2025-020 — Mira Volkov	Mira Volkov	Running	3	—	3m ago
INV-2025-019 — Northwind Holdings	northwind-holdings.io	Complete	3	→ case	3m ago

- **What:** the OSINT collection workspace.
- **Solves:** the iterative back-and-forth of building a profile.
- **Does not:** support bulk-edit on findings; per-agent re-runs aren't exposed (you can re-run the whole pipeline).

10.3 Pipeline

- **What:** the runner that drives the AI adapter end-to-end.
- **Solves:** turning a target into a structured set of findings with provenance.
- **Does not:** stream progress (stages animate in the UI but the server-side run is sequential per agent group).

10.4 Cases



- **What:** the forensic case manager.
- **Solves:** keeping evidence, branches, MRs, and assignments organised under one number.
- **Does not:** support file-byte upload (Evidence carries metadata + hash; bytes will land in Phase 8).

10.5 Evidence

- **What:** the cross-case evidence inventory.
- **Solves:** "find all evidence with hash starting 0ca . . . " / "show me every unverified item across all cases."
- **Does not:** pivot/group yet - it's a flat table.

10.6 Branch Graph

forenix-oss
OSINT • FORENSICS

FORENIX-OSS / Branch Graph

adapter mock v0.1.0 online Admin - admin sign out

Search cases, evidenc

OSINT

- Dashboard
- Investigations
- Entity Graph
- Monitors
- Verification
- Reports

PIPELINE

- Pipeline

FORENSICS

- Cases
- Evidence
- Network Graph
- Branch Graph**
- AI Lab
- Integrity
- Audit
- Reviews

ACCOUNT

- Teams
- Settings
- Admin

Branch graph — CASE-2025-007
Operation Sandstone

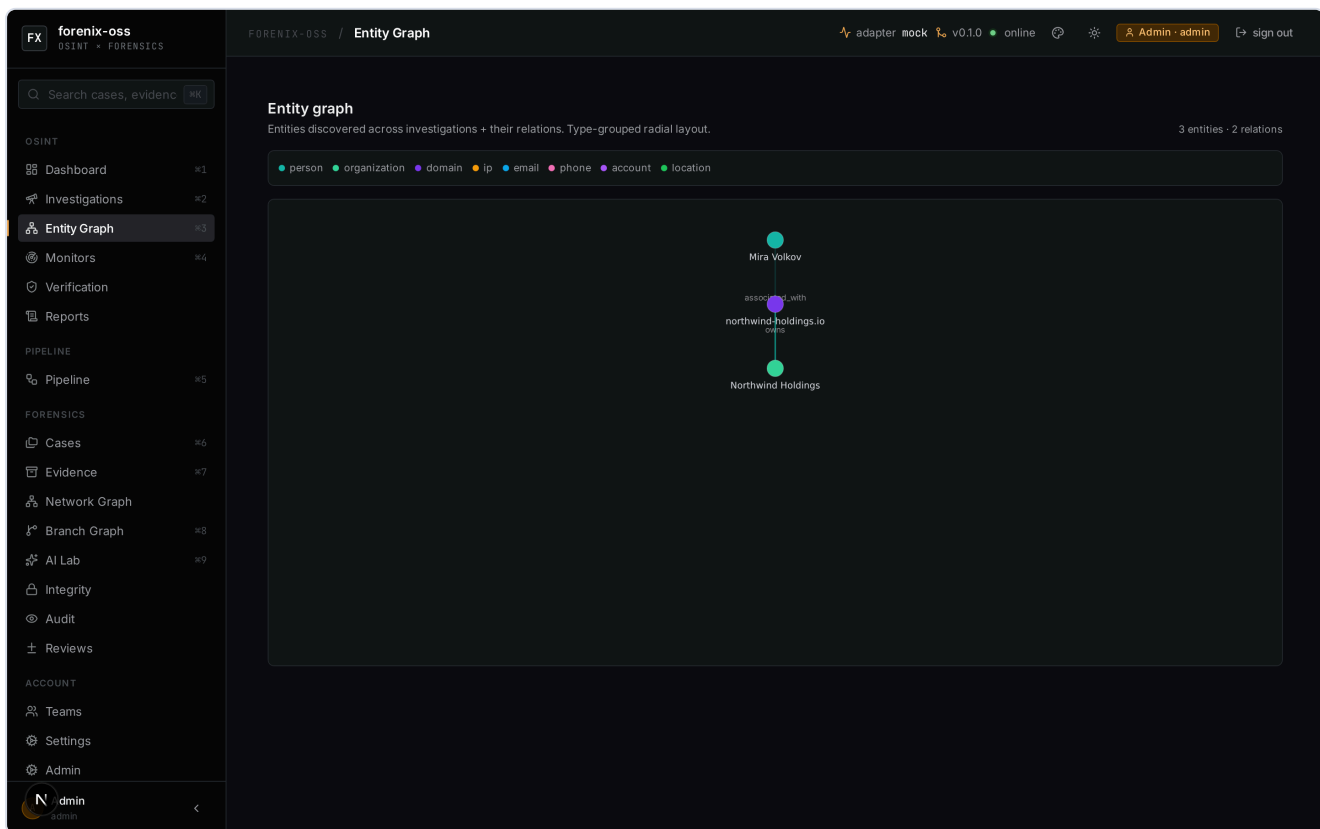
Pick another case

main 3 commits evidence-review 3 commits

ad7e385d..118a	ADD	add: smtp-headers-2025-05-02.eml	smtp-headers-2025-05-02.eml	unverified	4m ago
389fdd4b..964d	VERIFY	verify: chain of custody validated for smtp-headers-2025-05-02.eml	smtp-headers-2025-05-02.eml	verified	4m ago
07d63c2b..3ecb	ADD	add: whois-snapshot-northwind.json	whois-snapshot-northwind.json	unverified	4m ago
635b6c1..0420	VERIFY	verify: chain of custody validated for whois-snapshot-northwind.json	whois-snapshot-northwind.json	verified	4m ago
388c25d1..8767	ADD	add: pcap-handshake-fragment.pcap	pcap-handshake-fragment.pcap	unverified	4m ago
d3fa7230..4e0a	VERIFY	verify: chain of custody validated for pcap-handshake-fragment.pcap	pcap-handshake-fragment.pcap	verified	4m ago

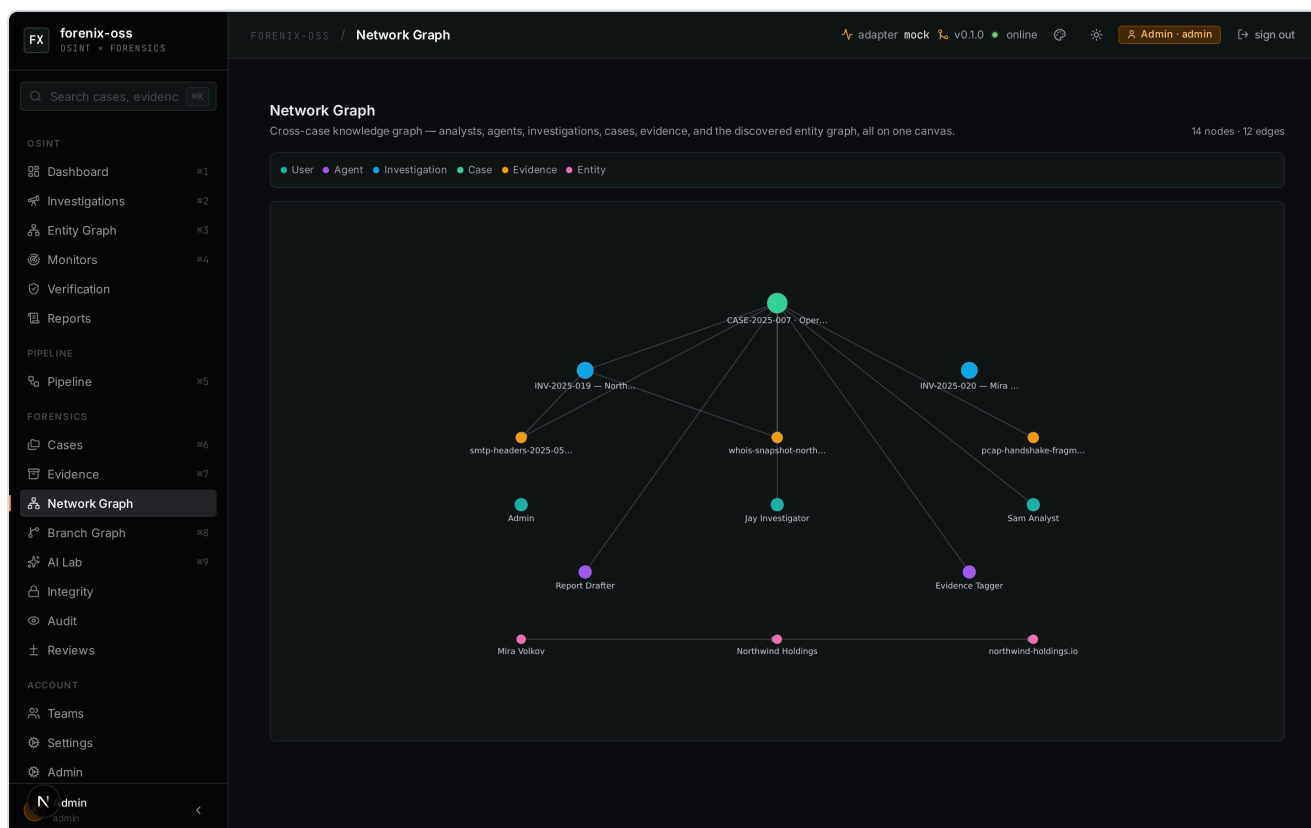
- **What:** Git-style commit graph for the active case.
- **Solves:** explaining the evidence-mutation history visually.
- **Does not:** support drag-to-reposition, no live diff view yet.

10.7 Entity Graph



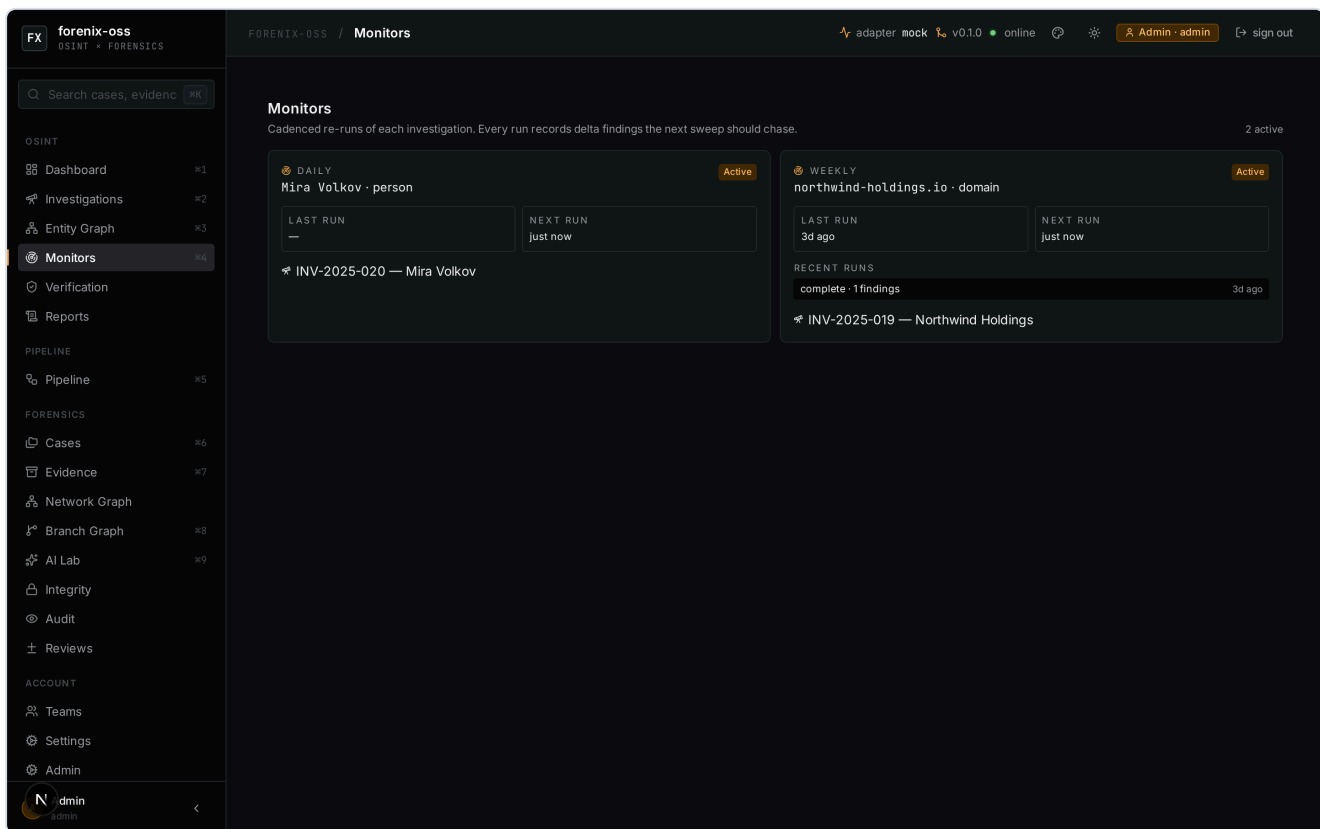
- **What:** OSINT entities + their relations laid out by type.
- **Solves:** "show me everyone we've discovered around this target."
- **Does not:** force-physics; deterministic radial layout only (which is why screenshots are stable).

10.8 Network Graph



- **What:** cross-case knowledge graph - users, agents, investigations, cases, evidence, entities, all on one canvas.
- **Solves:** seeing what connects two cases or two analysts.
- **Does not:** zoom / pan yet; labels overlap above ~250 nodes.

10.9 Monitors



forenix-oss
OSINT · FORENSICS

FORENIX-OSS / Monitors

adapter mock v0.1.0 online Admin · admin sign out

Monitors

Cadenced re-runs of each investigation. Every run records delta findings the next sweep should chase. 2 active

DAILY Active

Mira Volkov · person

LAST RUN	NEXT RUN
—	just now

INV-2025-020 — Mira Volkov

WEEKLY Active

northwind-holdings.io · domain

LAST RUN	NEXT RUN
3d ago	just now

RECENT RUNS

complete · 1 findings 3d ago

INV-2025-019 — Northwind Holdings

OSINT

- Dashboard #1
- Investigations #2
- Entity Graph #3
- Monitors #4**
- Verification
- Reports

PIPELINE

- Pipeline #5

FORENSICS

- Cases #6
- Evidence #7
- Network Graph #8
- Branch Graph #8
- AI Lab #9
- Integrity
- Audit
- Reviews

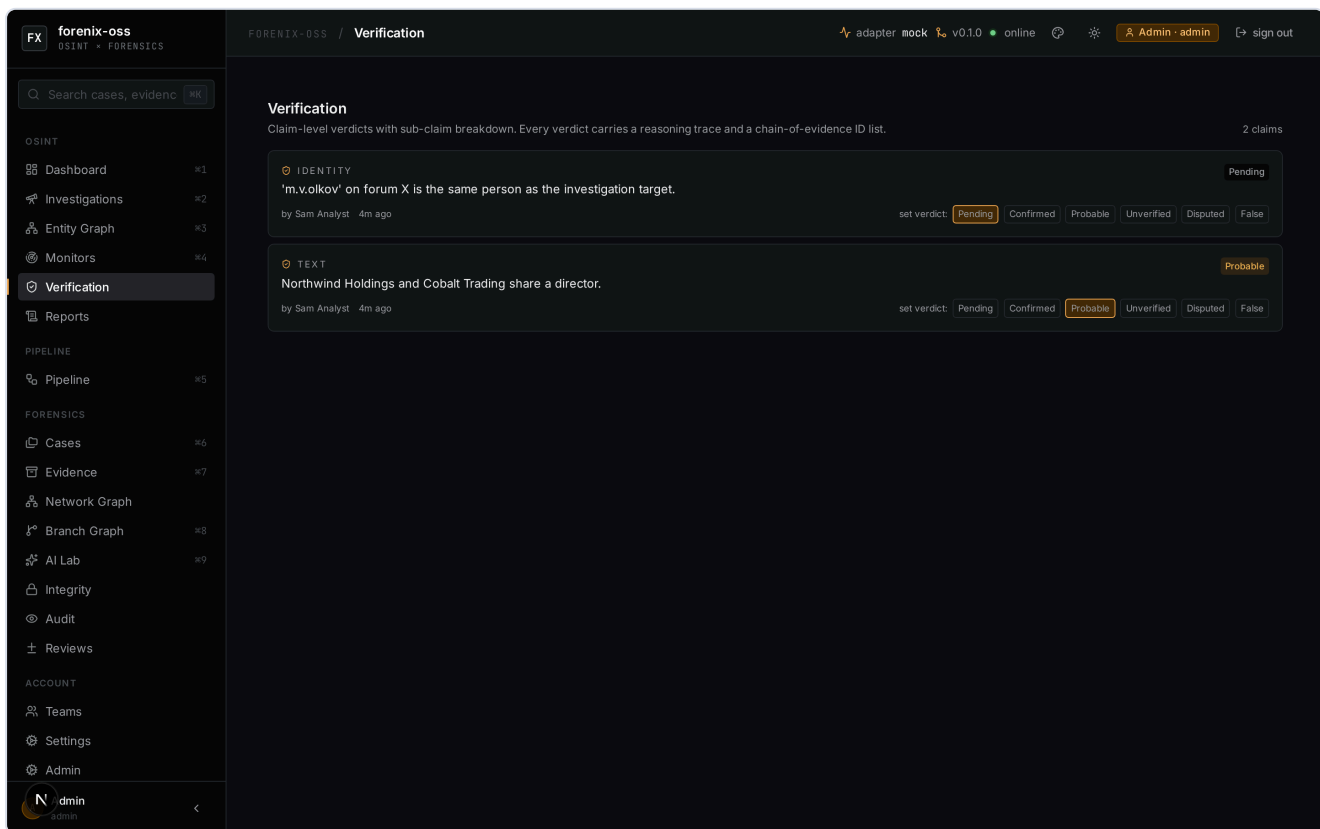
ACCOUNT

- Teams
- Settings
- Admin

N admin

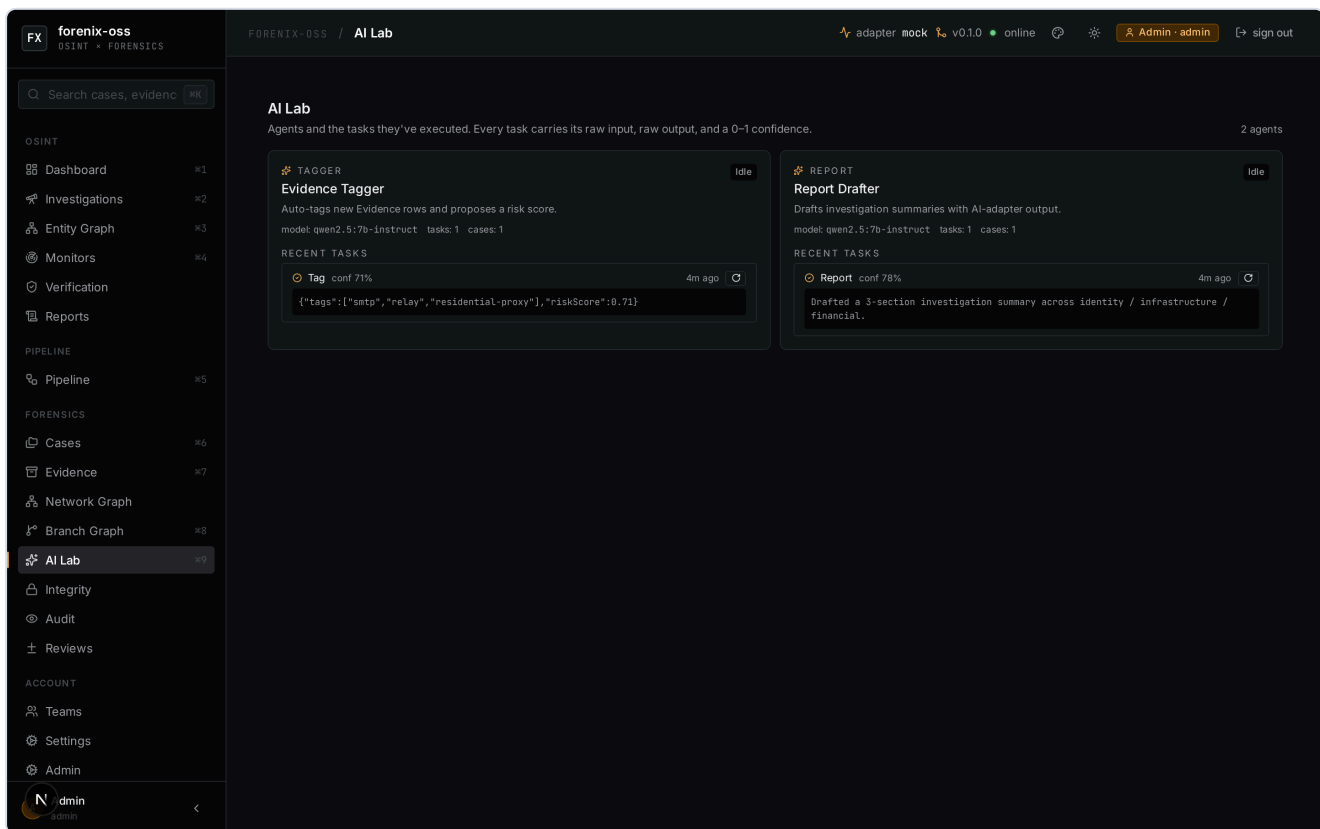
- **What:** cadenced re-runs of an investigation pipeline.
- **Solves:** keeping targets fresh after the report ships.
- **Does not:** actually wake the monitor - the background scheduler is Phase 7. The data layer is ready.

10.10 Verification



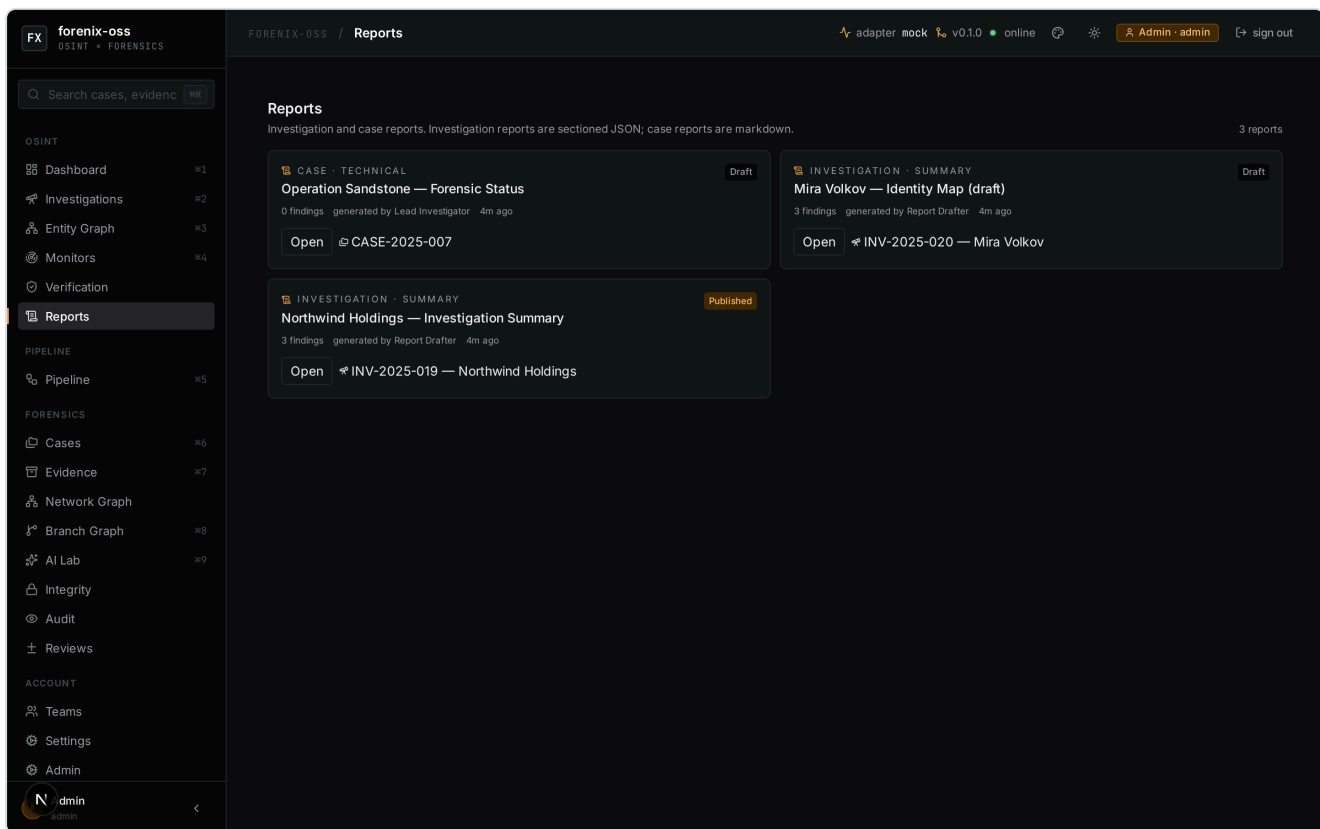
- **What:** claim-level verdicts (confirmed / probable / disputed / false).
- **Solves:** giving the human the final say over a model's confidence.
- **Does not:** support inline verdict editing yet (verdicts come from the API; UI mutations land next).

10.11 AI Lab



- **What:** what each agent was asked, what it returned, and how confident it was.
- **Solves:** debugging a misbehaving agent before its findings poison a case.
- **Does not:** support re-run / cancel from this page yet.

10.12 Reports



- **What:** both investigation-side and case-side reports.
- **Solves:** the deliverable to outside counsel / leadership.
- **Does not:** export to PDF (that's a SaaS-premium feature).

10.13 Audit

Audit log

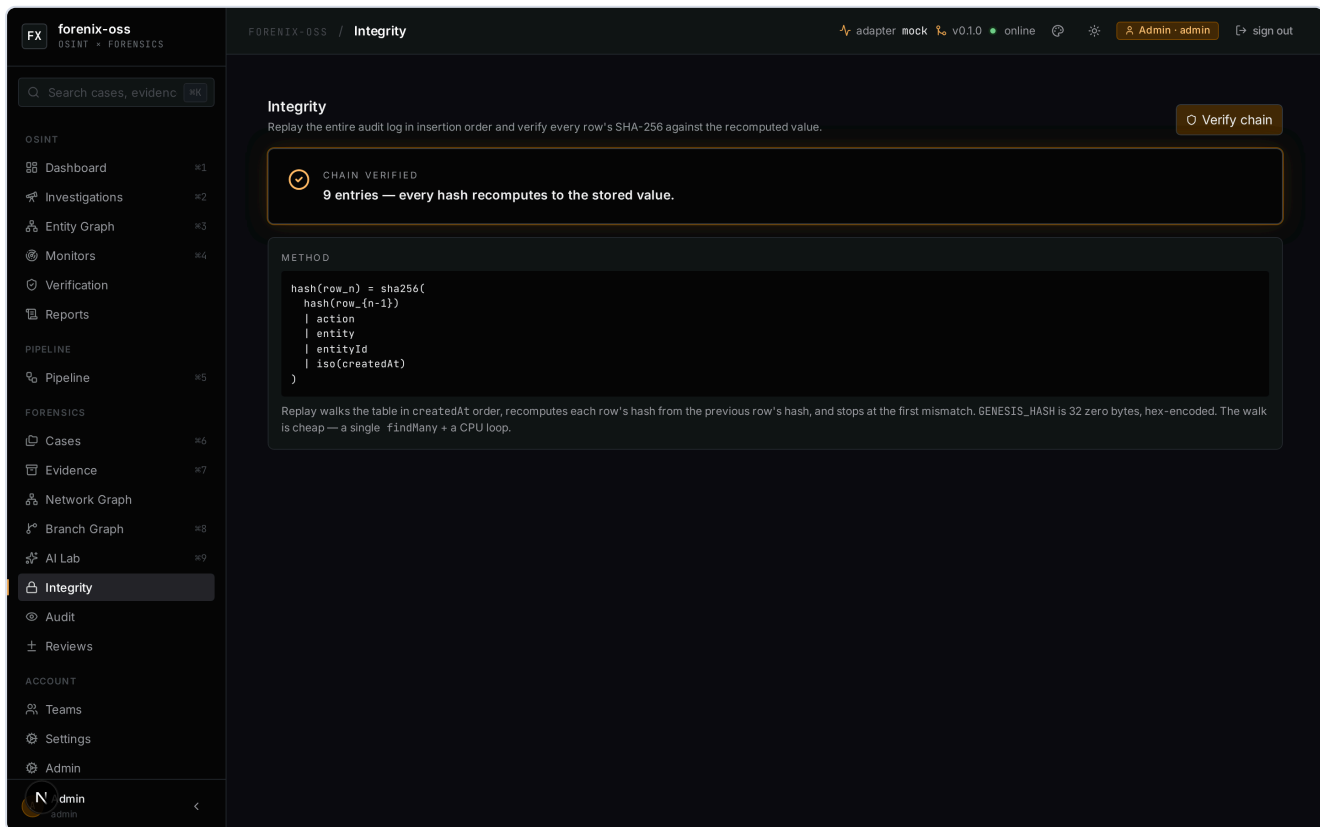
Append-only chain across both workflows. Every write computes sha256(prevHash | action | entity | entityid | iso(t)).

Q: action, entity, hash... 9 entries

HASH	ACTION	ENTITY	SCOPE	WHEN
4b465bca...7f26 .. 00000000...0000	create_user	User cmp97rsz8000	—	5m ago
bb10de21...a104 .. 4b465bca...7f26	create_user	User cmp97rt01000	—	5m ago
1b117e08...6856 .. bb10de21...a104	create_user	User cmp97rt05000	—	4m ago
61830660...082e .. 1b117e08...6856	create_investigation	Investigation cmp97rt2c001	inv	4m ago
52ad3c30...d86b .. 61830660...082e	run_pipeline	Investigation cmp97rt2c001	inv	4m ago
04361111...7fc1 .. 52ad3c30...d86b	create_case	Case cmp97rt0f000	case	4m ago
7003e873...d45b .. 04361111...7fc1	promote_to_evidence	Finding f-1	case	4m ago
7c363f93...3a5d .. 7003e873...d45b	verify_evidence	EvidenceCommit cmp97rt0r000	case	4m ago
bd73298b...44ab .. 7c363f93...3a5d	generate_report	Report r-1	inv	4m ago

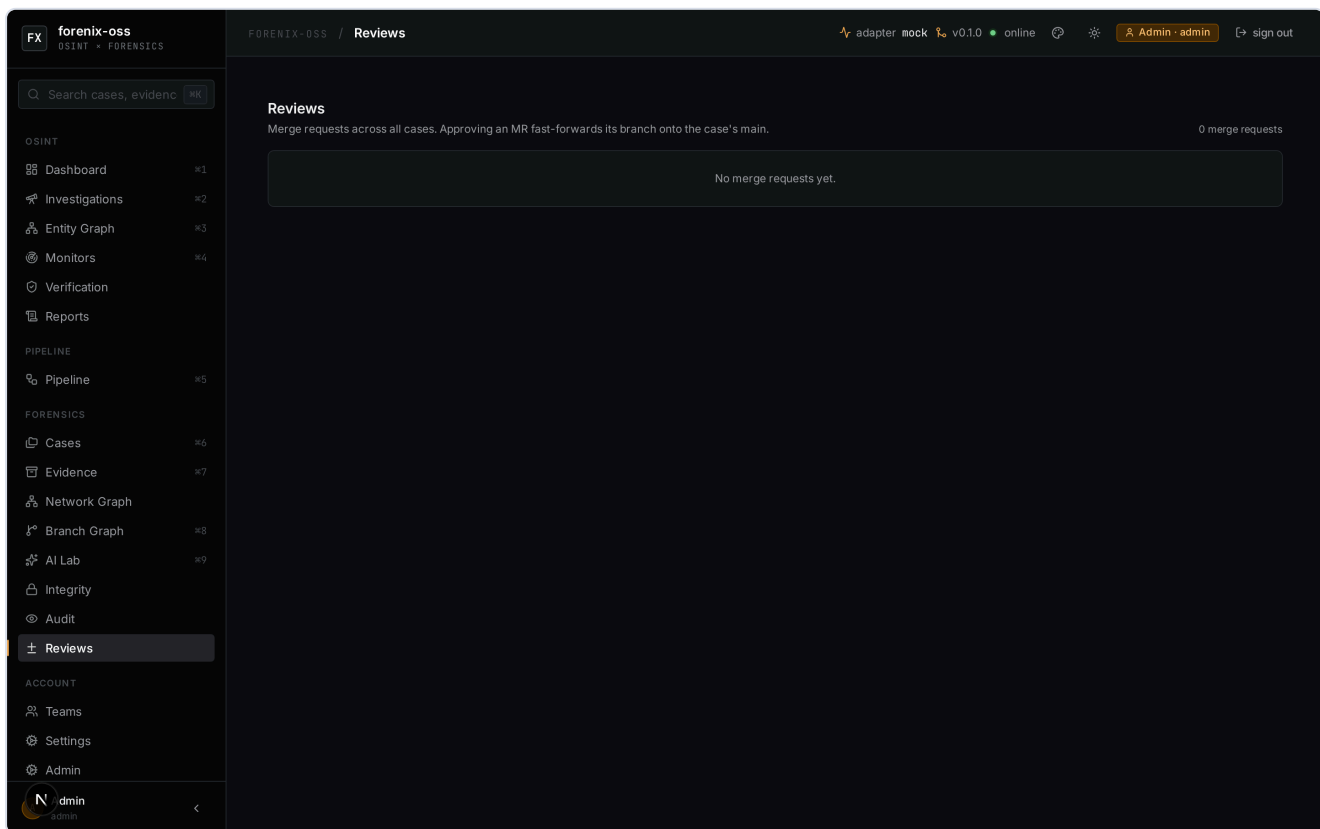
- **What:** the full hash-chained log.
- **Solves:** "who changed what, when, in what order."
- **Does not:** allow direct writes (the chain is computed by the system; you can't insert a row out-of-band).

10.14 Integrity



- **What:** one-button replay of the entire chain.
- **Solves:** proving tamper-evidence.
- **Does not:** use Merkle batching yet - linear replay is fast enough for sub-10M-row chains.

10.15 Reviews



- **What:** list of merge requests across all cases.
- **Solves:** second-pair-of-eyes review before evidence is sealed.
- **Does not:** offer inline merge/approve buttons yet - that ships in Phase 4 polish.

11. Working with the AI adapters

forenix-oss ships with six AI adapters. You can switch between them by editing one line in `.env` :

```
AI_ADAPTER=mock           # default - deterministic seeded output
# AI_ADAPTER=ollama        # local Ollama HTTP API
# AI_ADAPTER=glm           # Zhipu AI GLM-4 / GLM-5
# AI_ADAPTER=claude        # Anthropic Claude (SaaS-gated)
# AI_ADAPTER=openrouter    # OpenRouter proxy -> many models
# AI_ADAPTER=nvidia        # NVIDIA NIM hosted catalogue
```

After changing the value, restart the dev server. The top-bar adapter chip will reflect the change.

11.1 Per-call adapter override

You don't always want to commit to one adapter. Pass `adapter` in the pipeline-run body to override for a single run:

```
curl -X POST -H "content-type: application/json" \
  -d '{"agentGroups":["identity","social"],"adapter":"openrouter"}' \
  http://localhost:3000/api/pipeline/run/<INVESTIGATION_ID>
```

Use this when you want to A/B two models on the same target - say, NVIDIA's Llama-3.3-70B vs OpenRouter's `gpt-oss-120b:free` - and compare findings side-by-side.

11.2 Picking an adapter

Choose this	When
<code>mock</code>	demos, dev, screenshots, tests
<code>ollama</code>	air-gapped deployments / local-only workflows
<code>openrouter</code>	the most flexibility - one key, dozens of models
<code>nvidia</code>	high-quality 70B models on a free dev tier
<code>glm</code>	sovereign-China deployments / GLM-4 quality preference
<code>claude</code>	SaaS premium tier only

11.3 Costs + latency

Adapter	Cost (this build)	Latency (3-group run)
<code>mock</code>	\$0	< 1 s
<code>nvidia</code>	free dev tier	~ 45-60 s
<code>openrouter</code> (<code>gpt-oss-120b:free</code>)	\$0	~ 80 s
<code>openrouter</code> (paid models)	\$0.20 - \$5 per 1M tokens	15-30 s

The audit chain stays valid identically regardless of which adapter you use. The adapter swap only affects the *quality* and *latency* of the findings.

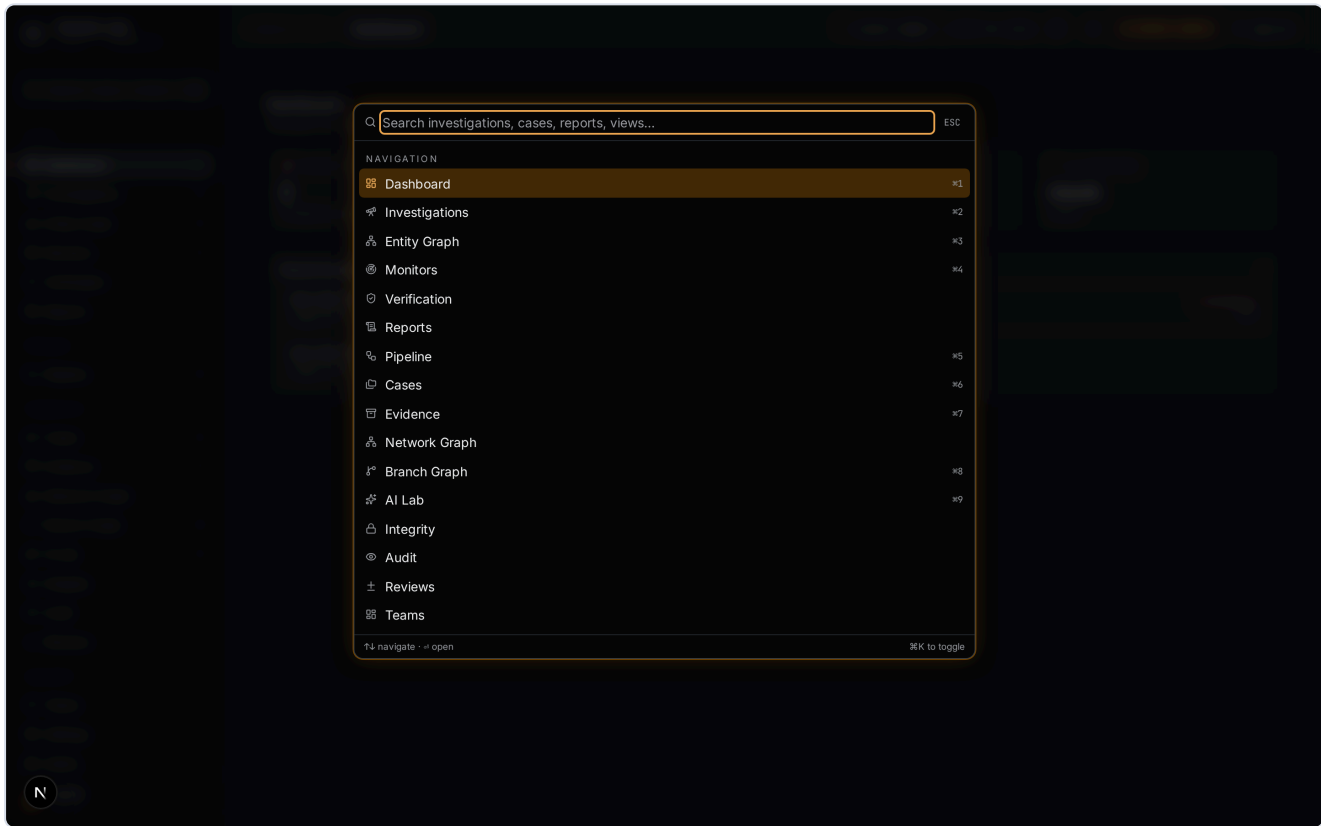
12. Keyboard shortcuts + command palette

12.1 View shortcuts

Shortcut	View
⌘1 / Ctrl+1	Dashboard
⌘2 / Ctrl+2	Investigations
⌘3 / Ctrl+3	Entity Graph
⌘4 / Ctrl+4	Monitors
⌘5 / Ctrl+5	Pipeline
⌘6 / Ctrl+6	Cases
⌘7 / Ctrl+7	Evidence
⌘8 / Ctrl+8	Branch Graph
⌘9 / Ctrl+9	AI Lab

12.2 The command palette

⌘K / Ctrl+K opens the palette from anywhere.



- Type to filter across nav items, investigations, cases, and reports.
- **^ / v** to navigate.

- ↵ **Enter** to open.
- **Esc** to close.

The palette is the fastest way to jump straight to a specific case or investigation by number/title.

12.3 Other useful shortcuts

- **Click any audit row's investigation/case scope chip** - switches view + activates the right detail panel in one click.
- **Click the bridge chip on an investigation** - opens its linked case.
- **Click a finding's *promote* -> *evidence* button** - does the promotion + writes the audit row + invalidates the queries that feed the surrounding UI.

13. Tips for new analysts

13.1 The verification habit

Make a habit of opening the Audit view at the end of every session and confirming `verifyAuditChain()` is still `{ ok: true }`. The check takes < 1 second and gives you the single most important guarantee the platform offers.

13.2 Promote sparingly

Only promote findings you genuinely intend to bring under chain of custody. A promoted finding becomes Evidence with a hash that will appear on every audit + court export. Use the verify-but- don't-promote pattern when you want to upgrade confidence but keep the item out of the forensic side.

13.3 Branch instead of overwriting

If you need to refine an evidence row's metadata, never edit it in-place. Create a new branch on the case, make the change there, open a Merge Request, and let a second pair of eyes review. This is the same workflow as Git, and for the same reason.

13.4 Filter before scrolling

Every list view supports an inline filter. Use it. The platform scales to thousands of investigations and tens of thousands of findings - scrolling does not.

13.5 Trust the chain

When the Integrity view turns green, it is genuinely green. The chain is cheap to verify (linear pass over a single table) and trivial to reproduce offline. Don't accept "trust us" answers from anyone - including us.

14. Troubleshooting

14.1 The dev server doesn't start

```
bun install      # re-install deps
bun run db:push  # re-apply the schema
bun run db:seed  # re-seed
bun run dev
```

If `bun` isn't installed, you can use `node` directly - `next dev` works the same. The seed script requires `tsx`, which is already in `devDependencies`.

14.2 The Audit Integrity view turns red

The chain has been mutated outside the platform. Do **not** try to "fix" it - the integrity guarantee depends on this being impossible to fix invisibly. Instead:

1. Note the `brokenAt` row id and timestamp.
2. Check `prisma/dev.db` (SQLite) or your Postgres for direct writes around that timestamp.
3. Escalate per your incident-response policy.

The chain remains broken until the next legitimate write - at which point the `brokenAt` row's `prevHash` is no longer the "tip" of the chain, but it remains visible to all future verifies.

14.3 A pipeline run hangs

Most hangs are slow upstream models. The platform aborts any single adapter call at 90 seconds.

- Switch to a faster model: `OPENROUTER_MODEL=openai/gpt-oss-120b:free`.
- Drop agent groups: try `identity` only first.
- Check the dev log: `tail -f /tmp/forenix-live.log` (or whatever you set as your log path).

14.4 The Pipeline runner shows "no investigations"

You haven't seeded the database. Run:

```
bun run db:seed
```

...or create one with the **+ New** button on the Investigations view.

14.5 "Pipeline 500" right after the run

The platform completed the run but failed to serialize the response (we've fixed every `BigInt` path in 0.1, but a future field might trip this again). The chain is still valid - refresh the investigation detail to see the findings landed.

15. Frequently asked questions

Does forenix-oss store any data outside my own database?

No. The platform is single-tenant. The only outbound traffic is to the configured AI adapter's API endpoint (you control that via `.env`).

Can I use a locally-hosted LLM?

Yes - the `ollama` adapter is built for exactly that. Pull a model with `ollama pull qwen2.5:7b-instruct`, set `AI_ADAPTER=ollama`, restart the dev server.

Can a non-technical team member use this?

Yes - the entire daily workflow is in the web UI. The only thing you need a terminal for is the initial install + the seed. After that, point your browser at `localhost:3000` and never look back.

How big can the audit chain get?

Tested cleanly to 100K rows on a single SQLite file. Beyond that, Postgres + the same algorithm gets you to tens of millions. We'll add Merkle batching when our first user hits the wall.

Will my findings stay private?

If you use a hosted adapter (NVIDIA / OpenRouter / Claude / GLM), the target name + search-result snippets are sent to that provider for analysis. Read each provider's data-handling policy. For zero outbound traffic, use `ollama` or `mock`.

Can I cite a forenix-oss case in a court filing?

The chain-of-custody methodology is documented and reproducible (see `docs/07-SECURITY.md`). Whether a court accepts it is a local matter - we'd love to hear from any user who runs this through their first admissibility argument.

16. Glossary

Term	Meaning
Adapter	A concrete implementation of the AI interface - <code>mock</code> , <code>ollama</code> , <code>glm</code> , <code>claude</code> , <code>openrouter</code> , <code>nvidia</code> . Pluggable, swappable with one env var.
Agent group	A bundle of OSINT agents that share a focus (e.g. <code>identity</code> , <code>infrastructure</code>). The pipeline runs the agents inside a group in parallel.
Audit chain	The SHA-256 forward chain over the <code>AuditLog</code> table. The trust anchor of the platform.
Branch	A named line of evidence-commit history under a case. Each case has a <code>main</code> branch by default.
Bridge	The act of linking an Investigation to a Case (via <code>Investigation.caseId</code>) and optionally promoting findings to Evidence.
Case	A forensic workspace. Owns Evidence, Branches, MRs, Assignments.
Commit	A versioned change to a piece of Evidence. Carries a SHA-256, a parent hash, an author, and a change-type.
Confidence	A finding's likelihood label: <code>confirmed</code> , <code>probable</code> , <code>unverified</code> , <code>disputed</code> , <code>false</code> .
Entity	A normalised actor in the OSINT graph - person, org, domain, IP, email, phone, account, location.
Evidence	A row in the forensic side of the schema. Carries a content hash + commit history. Can be <code>collected</code> / <code>verified</code> / <code>sealed</code> .
Finding	A discrete OSINT signal produced by an agent. Lives on the Investigation; can be promoted to Evidence.
Investigation	The OSINT collection workspace, bound to a <i>target</i> .
Merge Request	A request to merge one Branch into another, gated by review.
Monitor	A cadenced re-run schedule for an Investigation.
Pipeline	The end-to-end run that orchestrates <code>analyzePipeline</code> -> <code>extractEntities</code> -> <code>generateReport</code> over the AI adapter.
Promote	The act of turning a Finding into an Evidence row (one click in the UI).
Reasoning trace	The agent's free-text explanation of why it produced a finding. Stored on the Finding.
Report	A structured deliverable. Sectioned JSON (Argus-side) or markdown (ForenX-side).
Seal	Mark an Evidence row immutable. Writes a <code>seal</code> commit + audit row.
Target	The person / org / domain / IP / etc. an Investigation is about.
Verify (chain)	Replay every audit row's hash and confirm none have been mutated.

Term	Meaning
Verify (finding)	Flip a Finding's <code>verified</code> flag to <code>true</code> and record the verifier.

End of manual. For engineering / deployment / architecture references, see the documents under `docs/`.

If you spot something incorrect or unclear, please open an issue at <https://github.com/thunderstornX/forenix-oss/issues>.